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Climate Change, Agriculture, and Adaptation Options for Honduras

Arie Sanders, Timothy S. Thomas, Ana Rios, and Shahnila Dunston

Abstract

We use both biophysical and bioeconomic models to assess the impact of climate change on Honduran agriculture out to 2050. We find that for some key crops, such as maize and sugarcane, yield reductions will likely be larger in Honduras than most of the rest of the world will experience. We argue that the highest-value crop for Honduras—coffee—may also be the hardest hit by climate change.

Maize is projected to have a productivity loss of around 12 percent as a direct result of climate change, but because of increased prices from climate change, yields are projected to only decline by 9 percent, as farmers will invest more in productivity. Beans are projected to lose 10 percent in yield, even after adjusting for the increased investment in productivity by farmers.

Livestock may also experience productivity shocks due to climate change, particularly in the southern part of the country.

We make recommendations to policy makers to enact appropriate policies to help farmers adapt to the various productivity losses that would otherwise be experienced because of climate change.

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Introduction

Honduras is one of the poorest countries in the Latin America and Caribbean region. The World Bank classifies Honduras as a low-middle-income country. Per capita income in Honduras for 2013 was approximately \$2,180, but is very unequally distributed among its population.¹ Other social indicators, such as very high mortality rates for children under five years of age (22 per 1,000 children) and 65 percent of Hondurans living below the national poverty line, serve to reinforce the country's pressing economic needs (World Bank 2014).

Economic activity in rural areas is dominated by agricultural production, which accounts for 22 percent of gross domestic product (GDP) (BCH 2016). Some of the most important agricultural products are maize, coffee, bananas, and African palm, of which coffee and bananas are the largest export crops. Agriculture is the second-largest sector in Honduras, employing 40 percent of the labor force (Jansen et al. 2007).

Honduras is highly vulnerable to climate-related disasters (including hurricanes, tropical storms, and floods), and was the country most affected by extreme weather events during the 1996-2015 period (Kreft, Eckstein and Melchior 2016). Increasingly extreme climate conditions are anticipated (Schatan, Montiel, and Romero, 2010). Even by 2020, impacts of climate change are expected to be equivalent to at least 0.17% of GDP by 2020 (CEPAL, 2011). Projected impacts from climate change include rising sea levels, stronger tropical cyclones, altered rainfall, storm surges, and increasing temperatures (World Bank 2009). Vulnerability to climate change is exacerbated by the lack of effective governance structures, high rates of population growth, and urbanization, as well as by poor land-use planning, which results in environmental degradation and habitat destruction (Sanders et al. 2015).

In this report, we assess the potential impacts of climate change through 2050 on the agriculture, forest, and land-use (AFOLU) sectors of Honduras, using various modeling tools. We evaluate the direct impacts without considering changes in GDP, population, agricultural technology, or adaptation options. Then, using the findings, we assess the potential impacts, taking the other projected changes into consideration, along with climate change. To that extent, this report can be seen as a longer-range planning tool that is more useful than simply considering the impacts of climate change alone, because it considers resources and challenges that are likely to be part of the environment policy makers will face in the future.

Most of what has been written to date on the impacts of climate change on agriculture in Honduras has been based on the climate models from the Intergovernmental Panel on Climate Change's (IPCC's) Fourth Assessment Report (AR4, IPCC 2007), or even earlier. In this report, we use the climate models from the Fifth Assessment Report (AR5, IPCC 2014). These are the latest and best models that are available on a global scale. They are used as inputs into crop models, as well as in the International Model for Policy Analysis of Agricultural Commodities and Trade (IMPACT) global partial equilibrium model and AR5's economic and demographic assumptions for the shared socioeconomic pathways (SSPs).

This discussion paper firsts presents the economic and demographic trends of Honduras. It then describes the climate, in terms of both what Honduras experienced in the second half of the 20th century, and what is projected for Honduras through the middle of the 21st century. The next section examines the current situation for the AFOLU sectors, and then explains what the models suggest will happen to them with

¹ All currency is in current US dollars unless otherwise indicated.

climate change. As part of this analysis, we make recommendations for measures to help farmers adapt to climate change.

Geographic and Agroclimatic Information

Honduras is in Central America, bordering the Caribbean Sea and the northern Pacific Ocean, with Guatemala located to the west, Nicaragua to the southeast, and El Salvador to the southwest. It covers a total area of 112,000 square kilometers (km²) and is the second-largest country in Central America after Nicaragua. Honduras has three distinct topographical regions: an extensive interior highland area and two narrow coastlands (Figure 1).

The interior highland is a vast interior plain with hills and valleys that make up more than 80 percent of the landscape. The area is considered to have lower agricultural potential, with slopes averaging more than 30 degrees and thin, rocky soils (Southworth, Munroe, and Nagendra 2004). However, scattered throughout the interior are also numerous flat-floored valleys that are 300–900 meters above sea level and provide sufficient grass, shrubs, and dry woodland to support livestock. In some areas, such as the Comayagua department, these valleys can support commercial agriculture (Merrill 1995).

The Caribbean lowlands in the North consist of river valleys and coastal plains. The broadest river valley along the Río Ulúa is Honduras' most developed area. San Pedro Sula, Honduras' second-largest city, is in this valley, along with a high concentration of agricultural activities. Located in approximately the center of the north coast of Honduras is the city La Ceiba, which is highly vulnerable to floods and sea level rise because of its topography (Smith et al. 2011). To the east, near the Nicaraguan border, the Caribbean lowlands broaden to an extensive region known as the La Mosquitia. This area consists of inland savanna with swamps and mangrove forests near the coast (Harborne, Afzal, and Andrews 2001).

The last and smallest physiographic region of Honduras is the Pacific lowlands. This strip of land averages 25 km wide on the north shore of the Gulf of Fonseca, and 185 km of the gulf's 261-km coastline are in Honduras. The land is flat, becoming swampy near the shores of the gulf, and is composed mostly of alluvial soils washed down from the mountains (Merrill 1995). The gulf itself has a variable depth, but is characterized by shallow waters. The Gulf of Fonseca is considered one of the most important tropical coastal systems along the eastern Pacific Ocean because of the size of the estuarine complex and mangrove belt (GEF and IADB 2006).

Economic and Demographic Conditions and Projections

Climate change is only one of the important trends that will affect the well-being of Honduras by 2050. At the same time the climate is changing, GDP and the population will be growing, with the latter most likely growing at a more modest rate.

Population

Past and Present

Honduras is one of the poorest countries in Latin America, with almost 60% of its population living in poverty (World Bank 2015). Poverty rates are higher among rural and indigenous people and in the South and West and along the eastern border than in the northern and central areas where most of the nation's

industries and infrastructure are concentrated (World Bank 2006). The increased productivity needed to break Honduras' persistent high poverty rate depends, in part, on further improvements in educational attainment. Although primary school enrollment is high, 6 percent of children of official primary school age are not in school, and the educational quality of public schools is generally poor (EPDC 2014). Honduras' population growth rate has slowed since the 1990s, but it remains high at nearly 2 percent annually, with the birth rate at approximately three children per woman and more among rural, indigenous, and poor women (UN 2011). Consequently, Honduras' young adult population—ages 15 to 29—is projected to continue to grow rapidly for the next three decades and then stabilize or slowly shrink (PNUD 2012). Population growth and limited job prospects outside of agriculture will continue to drive emigration to the United States (Reichman 2013). In 2015, remittances from Hondurans working abroad represented about a fifth of Honduran GDP (BCH 2016).

Figure 2 shows demographic trends for the last four decades. The demographic trends reveal two important conclusions:

1. Population growth is high during the entire period—an average of 2.5 percent. While the growth rate has slowed over the last two decades (to an average of 2 percent), the country is growing slowly, with a fertility rate of three births per woman. Nevertheless, this is more than the replacement rate (UN 2015).
2. The proportion of the population living in urban areas has been steadily on the rise, and was approximately 55 percent by 2015. During the last five years, the urban population increased by 3.1 percent. As can be expected in a country with a limited budget, the urban infrastructure has not kept up with the rapid growth, which has led to the creation of unplanned settlement in the hills around Tegucigalpa (PNUD 2012).

Figure 3 shows the distribution of the population throughout the country. Most of the population lives in the western half of the country, with most of the concentration located in the urban areas. The Río Plátano Biosphere Reserve is in the far eastern half of Honduras. The reserve is one of the few remains of a tropical rainforest in Central America, and because of its classification as a biosphere reserve and its lack of connectivity, it is among the least populated areas of the country.

Future

In light of the historical population trends, IPCC demographic projections show continuous growth of the Honduran population (Figure 4). However, there are significant differences in the slope of the curves. In the figure, shared socioeconomic pathway (SSP)3 and SSP2 have a very similar path of higher economic development, while SSP1 is a slower economic development scenario. SSP3 indicates that in 2050 the population will be doubled, while SSP1 expects a demographic stabilization at around 9 million inhabitants for the same year. Since the economic modeling in this report assumes SSP2, we should also make special note that under this scenario, population is to rise to around 9 million through 2030, and at 11 million by 2050. The urban population as a percentage of the entire population is expected to further increase (Figure 5). By 2040, SSP3 estimates that the urban population will stabilize at around 55 percent, SSP2 estimates it will be above 60 percent, and SSP1 estimates it will be above 70 percent.

Economic Development and Vulnerability

Past and Present

Figure 6 shows two trends important to a discussion of well-being and poverty. While GDP per capita growth was favorable in the 1970s, this trend turned negative in the 1980s because of the political turmoil in the region, which had strong repercussions for the Honduran economy. During the 1990s, Honduras faced different kinds of environmental and economic shocks. In 1998, Hurricane Mitch damaged or destroyed tens of thousands of homes, washing away at least 70 percent of the crops, including 80 percent of the banana crop. The total damage was estimated at more than \$5 billion (CEPAL 1999). During the end of the 1990s, coffee prices in the international market decreased because of the continued increases in world supply. Honduras registered an approximate 33 percent decline in export revenues from crop years 1999/2000–2000/2001 (Varangis et al. 2003).

Between 2005 and 2014, Honduras experienced steady and notable economic growth, with GDP per capita being around one-third higher at the end of the period (\$2,190) than at the beginning of the period (\$1,380) (World Bank 2014). However, GDP per capita does not tell a complete story about the well-being of the population. Table 1 presents additional indicators of well-being, which indicate reasonably high levels, with secondary school enrollment rates, adult literacy, and access to electricity being moderately high, and malnutrition being moderately low.

Figure 7 shows improvements in two indicators of well-being between 1970 and 2012: (1) the under-five mortality rate, which dramatically fell from around 150 per thousand to less than 30 per thousand; and (2) life expectancy at birth, which rose from less than 60 years to 73 years.

The quality and coverage of infrastructure and transport services have a major impact on living standards and economic growth. Figure 8 shows travel time maps for Honduras, which are meant to give rough indicators of connectivity for the agricultural sector. Particularly important is whether farming areas are sufficiently connected to markets to be able to sell their produce and acquire inputs like seeds and fertilizer. Even connection to towns as small as 20,000 can be vital for farmers, since such towns can readily serve to interact with farmers' buying and selling needs, in addition to consumer goods.

There is a marked difference in road access between the eastern and western parts of the country. Road connection is fairly good throughout the central-western part of Honduras, including access to some small towns. However, connectivity can be a two-edged sword. On one hand, it facilitates the economic development of those who live there, but on the other hand, as the development is facilitated, it tends to attract more people. In the case of protected forested areas, increased connectivity is not necessarily an advantage.

Future

The projected economic growth paths for Honduras in Figure 9 present significant differences. Under the SSP3 and SSP4 scenarios, economic growth per capita would be almost flat. Both scenarios are similar to the scenario of the 1990s, where population growth equals the increase of GDP and negatively affects Honduras' strategy to reduce its poverty rates. In the more optimistic scenarios of SSP1 and SSP5, GDP per capita increases almost fivefold. Economic growth is an important precondition for poverty reduction. The fact that Honduras accomplished the first Millennium Development Goal—to eradicate extreme poverty and hunger—was largely the result of its favorable economic growth rates.

Climate and Climate Change

Current Climate

The Global Climate Risk Index (Harmeling and Eckstein 2013) analyzes how extensively countries have been affected by the impacts of weather-related events, such as storms, floods, and heat waves. The 2013 index confirms that less-developed countries are generally more affected than industrialized countries, in terms of both being exposed to increased risks and being less resilient to climate hazards. According to this analysis, Central America is particularly vulnerable to the negative impacts of extreme climate events. More specifically, for the period 1993–2012, Honduras was ranked as the most vulnerable country in the world.

The table and figures in this section look at the climate of the 1960–1990 period, which we use as a baseline for changes (Hijmans et al. 2005). The climate of Honduras is shaped by its geographical location, with the north coast bordering the Caribbean, the south coast bordering the Pacific Ocean, and the orientation of the main mountain chains influencing the direction of the trade winds. The distribution of rainfall is highly variable from North to South, with extremely hot and humid areas along the Caribbean coast; temperate and rainy zones as in the high peaks of the mountains; and dry, warm climates with less than 500 millimeters (mm) of total annual average rainfall, as in some areas of the South (PNUD 2010).

The cold front season, which usually occurs between November and March, leads to heavy rainfall along the northern coast, and tropical cyclones strike between June and November (Kawas 2012). Moreover, the availability of water is altered by the phenomenon of El Niño (and alternatively La Niña), which has caused frequent incidents of torrential rains with heavy flooding. Storms and floods have had high human and economic impacts on Honduras. In addition to the damage caused by Hurricane Mitch in 1998 (which was already discussed), between 1999 and 2008, storms affected more than 200 thousand people and caused \$127 million in damages, and floods affected 15,000 people with an estimated damage of \$128 million (World Bank 2009). The most severe droughts have been concentrated in the departments of Choluteca, Valle, La Paz, El Paraíso, Francisco Morazán, Intibucá, and Lempira (Ordaz et al. 2010).

The Caribbean lowlands, in the Northeast, are generally rainier than rest of the country, with approximately 2,600 mm of precipitation annually (Figure 10). Located in this region is the Río Plátano Biosphere Reserve, the largest area of concentrate forest in Honduras, covering 5,252 km² or nearly 7 percent of the national territory (Stevens et al. 1997). In the Northeast, rainfall is sufficient to support cultivation, but may become excessive and inhibit plowing and other farming activities (Porch et al. 2007).

In contrast, the South and West are part of the “dry corridor,” where rainfall is scarce and not sufficient to support cultivation, plowing, and other farming activities. Here, the original forests have undergone almost total conversion, leaving only patches of secondary forest in the foothills (Gordon et al. 2003).

Figure 11 shows the rainfall map for the wettest three consecutive months of the year—the approximate precipitation that would occur during a growing season—while Figure 12 shows the rainfall in the driest three consecutive months of the year, with the driest areas in the dry corridor.

The map in Figure 13 shows the mean daily maximum temperature for the warmest month of the year. As expected, the points with the highest elevation report lower temperatures and represent the main mountain chains of Honduras. Also, the Caribbean and Pacific coastal areas have the highest temperatures

in the country during the warmest month. Table 2 summarizes the statistics presented in Figures 10 through 13.

Future Climate

Potential climate impacts for Central America include increased risks of food insecurity and famine, increased water stress and water availability problems, disruption of coastal marine resources, threats to human health, damage to infrastructure, greater vulnerability to and risk of disasters, and threatened livelihoods and culture of indigenous people (Estado de la Nación 2011).

Figure 14 presents projected changes in annual precipitation for the period 2000–2050. There is a significant discrepancy among the climate models. The Geophysical Fluid Dynamics Laboratory (GFDL) model projects considerable rainfall increases, particularly in the Northeast and West, while the L’Institut Pierre-Simon Laplace (IPSL) model predicts a severe reduction in rainfall, especially in the South and Southwest. The other two models show a moderate reduction in annual precipitation, though the Hadley Centre Global Environmental Model (HadGEM) projects the smallest reduction in rainfall among the four models.

Figure 15 shows projected changes in the mean daily maximum temperature for the warmest month of the year, with an increase of 1–3 degrees Celsius (°C) in the actual temperature. The GFDL model shows a smaller increase, with small patches, especially in the South. In contrast, the HadGEM projected increases of around 3°C or higher in some areas of the West. The other two models show very similar projections for increasing temperature: the IPSL model exhibits an equal increase of 2.3 °C for all of Honduras, and the Model for Interdisciplinary Research on Climate (MIROC) projected the West as the hottest region.

Table 3 presents the changes in annual precipitation and mean daily maximum temperature projected by general circulation model (GCM) for the period 2000–2050. The GFDL model’s results for annual rainfall differ significantly from those of the other models, with the GDFL projecting an increase of 521 mm, and the other three projecting decreases of 98–352 mm. But all four models forecast a general increase in temperature in the country of 1.89–2.78 °C, especially in the South and Southwest.

Government Policy

The implementation of the country’s National Climate Change Strategy resulted from the international commitments Honduras made by signing and ratifying the United Nations Framework Convention on Climate Change (UNFCCC) (SERNA 2012). The UNFCCC constitutes the framework for establishing a national policy for climate change, as well as defining and implementing the most appropriate strategies for adaptation and mitigation. This UNFCCC includes 17 strategic objectives that are classified in seven areas for action: water resources; agriculture, soil, and food security; forests and biodiversity; coastal and marine ecosystems; human health; risk management; and hydroelectricity.

Honduras has submitted two UNFCCC national communications (SERNA 2000, 2012). The first, published in 2000, reported the actions taken by the Honduran government and the analytical basis of its policy response to climate change. The second communication, submitted to the UNFCCC in 2012, includes the country’s latest information on climate change mitigation and adaptation actions at the local level.

Honduras has made important progress in the establishment of a normative and institutional framework to tackle climate change. Examples of such efforts include the climate change law and the national climate

change strategy.² Additionally, in 2014, the Secretary of Agriculture and Livestock (SAG) published the country's national strategy for efficiently adapting its food and agriculture sector to the effects of climate change and simultaneously developing the sector sustainably (SAG 2014). This goal is in line with the country's National Climate Change Strategy and other sectoral policies (SAG 2014). Assessment of the actual and potential impacts of climate change and related adaptation strategies revealed that this small-scale sector is the most vulnerable. Therefore, Honduras has ranked agriculture and food security as its first priority for adaptation, since agriculture affects the majority of the country's rural population's livelihoods and food security. Several activities have been proposed by SAG on agriculture and food security, including the promotion of micro-irrigation, drought-tolerant crops, and climate change awareness among farmers.

Honduras' Nationally Determined Contribution (NDC) establishes a 15% reduction in greenhouse gas emissions by 2030, and identifies priority sectors for mitigation and adaptation (Gobierno de Honduras, 2015)³. Agriculture is a priority for mitigation and adaptation. Additional key adaptation actions are focused on land use, forestry and food security.

So far, measures to adapt to the effects of climate change, particularly related to the agriculture sector and livestock subsector, are designed to optimize production processes. The strategies outlined in the National Plan of Action on Climate Change seek to influence the reduction of emissions of carbon dioxide and methane and thus prevent and control land degradation (Ordaz et al. 2010; SERNA 2011).

Impact of Climate Change on Agriculture, Forests, and Land Use

Figure 16 shows the main land cover in each grid cell of approximately 1 km on edge. We note that the predominant land cover type is tree cover, followed by cropland and grassland. In the following sections, we delve more deeply into the distribution of crops, livestock, and forestry.

Agriculture: Overview

Crops

Honduras is characterized by duality, where a modern large-scale agricultural production system coexists with a smallholder, resource-poor, subsistence, and semi-subsistence production system. Historically, land ownership patterns throughout Honduras have meant that large landowners own the majority of the most productive land in the central and northern valleys, situated on flat land with easy access to groundwater for irrigation. In contrast, smaller, poorer agricultural producers work the hillside land, which is characterized by poor soils (Jansen et al. 2007). Large-scale farmers also have better access to technology and financial markets.

The large-scale system produces high-value crops for export, providing a large source of export earnings. Historically, Honduras' key export crops have been bananas and coffee. Today, however, African palm oil,

² The climate change law and the national climate change strategy were approved in 2014 and 2010 respectively (La Gaceta, 2014).

³ Honduras ratified the Paris Agreement on September 21, 2016. Hence, the Intended Nationally Determined Contribution (submitted to the UNFCCC in 2015) turned into its Nationally Determined Contribution.

produced in the country's northern and central regions, is becoming an important generator of export revenue, followed by sugarcane and, to a lesser extent, pineapple and melon (BCH 2015).

More than 85,000 small producers (selling fewer than 80 bags) collectively grow more than 90 percent of all coffee produced in Honduras (Gomez 2012). Motivated by the high international price of coffee, during the last five years more than 10,000 new coffee producers entered the market. Many of those small farmers converted their land holdings to coffee production. Coffee production in 2010/2011 accounts for 40 percent of total agricultural exports (Banegas Barahona et al. 2012), and the 2010/2011 sales contributed to 27 percent of the country's agricultural GDP (USDA FAS 2012). As many households that engage in the coffee sector work as wage laborers (more than 1 million people), the nearly 30 percent decrease in demand because of coffee leaf rust during the last three years has reduced their incomes and compromised their ability to meet their daily food needs (FEWS NET 2014).

On the other hand, the subsistence sector, which produces maize and beans (called basic grains in the Central American context), is mostly made up of small farms with minimal access to markets, inputs, and improved seeds (Serna 2007). Except for large-scale maize production in western Honduras, maize and beans are mostly rainfed, which make farmers vulnerable to climate variability and long-term climate trends. These basic grains are primarily grown for subsistence, although there is some commercial production as well, especially beans. Seasonal hunger tends to occur in rural areas and follows the seasonal rains (Jansen et al. 2007).

Crop production is characterized by low productivity levels. While yields in the large-scale system are similar to regional averages, maize yields at the subsistence level are among the lowest in the region—approximately 1.5 metric tons per hectare (MT/ha) (Hintze, Renkow, and Sain 2003). Moreover, corn yields are 30% and 20% lower than those in Guatemala and El Salvador respectively⁴ Increased production of basic grains has mainly been achieved by expanding the area of land under cultivation. Low fertility and soil erosion are serious problems in the “milpa system” (intercropping of maize and beans), primarily caused by hillside farming and intensive cultivation of land without the use of sufficient fertilizers (Jansen 2007). Also, most of the countless number of agricultural development projects designed to promote the introduction of conservation technologies, such as cover crops, have not been very successful in achieving their objectives (Neil and Lee 2001). While cropland is mainly found in the central and western regions of the county, the cropland area is denser to the South, which includes the departments of Valle and Choluteca (Figure 17).

Table 4 summarizes recent agricultural production for Honduras. The logic behind using land use as the measure for establishing the order of crop importance is that whatever farmers value enough to devote their land to must be the item of greatest importance to them, and in a similar sense, perhaps should be thought of as having the greatest value to the nation's agriculture sector. The table shows that the five leading crops are maize, green coffee, dry beans, palm fruit oil, and sugarcane. The area of land harvested declines after the fourth-ranked crop. Maize is planted on more than one-third of the area that is cultivated, coffee on almost 23 percent, beans on almost 11 percent, sorghum on 9 percent, and sugarcane on almost 6 percent.

Over the last decade, planting area has been mostly stable, except for palm oil, which has expanded by around 181 percent (Figure 18). In the case of northern Honduras, palm oil expansion is the most

⁴ Estimation based on FAOSTAT 2010-2014 yield data.

important proximate cause of land-use change and has been the primary driver for converting natural areas (directly or indirectly) to land for agriculture (Sanders, McLean, and Manueles 2015).

Apart from year-to-year variation, variation in yields has been small (Figure 19). However, coffee yields have risen by around 50 percent, which can be partly explained by the reduction of illegal coffee export to Guatemala. Guatemalan coffee is considered to be premium coffee and has received a higher price for its quality than Honduran coffee; however, an increase in the quality of Honduran coffee has reduced the price difference between both countries.

Livestock

The livestock subsector represents 35 percent of the agriculture sector and employs 13 percent of economically active Hondurans (Pérez et al. 2006). Beef exports to the United States, although historically important in the growth of the subsector, currently account for a minor part of the country's overall economic activity. Cattle contributed 9.1 percent to agricultural GDP. In 2010, and 10.9 percent in 2014 (BCH 2014). Nonetheless, the livestock sector is characterized by inferior productivity and low technology (Pérez Destephen 2012).

Livestock accounts for one of the most important land uses in Honduras, with 1.5 million ha dedicated to pasture (ICF 2014). In 2008, small producers controlled 52 percent of the nation's pastureland (10–50 ha) and managed about 56 percent of the cattle. Of a total of 97,000 cattle farms, about 10,000 own 92 percent of the pastureland. Between 1993 and 1999, the number of cattle decreased from 2.8 to 1.7 million in 1999 as a result of both increased exports to Guatemala and Mexico (Pérez et al. 2006) and the destruction caused by Hurricane Mitch in 1998. During the past few years, the number of cattle has grown at a moderate rate, which can be explained to a general change in consumer demand from beef to less expensive proteins, such as poultry and pork (USDA FAS 2017).

Milk production varies depending on the season. For Honduras, milk production increases during the winter, or the rainy season when more pasture is available, and decreases during the summer, or the dry season. During the summer of 2008, approximately 470,000 cows produced 1.79 million liters of milk per day, for an average yield of 3.8 liters/cow/day. During the rainy season of 2008, the number of cows increased by 90,000, producing an average of 2.44 million liters of milk per day, for an average yield of 4.4 liters/cow/day (INE 2008).

Pig production is a traditional activity of great importance for the agriculture sector. The pig population in Honduras has remained stable since 1993, with slight variations between periods. In 2008, the country reported a total pig population of 450,000 animals. The stagnation in the population is explained by the lack of high-quality breeding sows and the limited availability of complete foods that allow for efficient meat production. Further, many animals are exported to El Salvador, which leaves a great void in the ability to meet domestic demand. As a result, meat imports, especially from the United States, have increased (Pérez et al. 2006).

Regarding poultry production, historically, Honduras has been an importer of chicken meat and eggs. In the early 2000s, the government focused on changing this trend to protect the national heritage. Protection programs have been implemented successfully to incentive domestic production. However, this specific policy is a classic example how protection may stimulate market inefficiency. Current poultry meat price for end consumers in Honduras is 56% higher compared to international prices (Shik et al. 2016). Also, in rural areas, small farmers use chickens and pigs as a way of saving money, and can sell or

eat the animals during periods of economic stress (Jansen 2007). Table 5 shows a count of livestock in Honduras, averaged between 2010 and 2012.

Figure 20 shows a steady rise of the number of chickens, with the population more than tripling between 1990 and 2012. The cattle population also increased in the same period, but at a steadier pace. In contrast, the pig population shows a reduction between 1990 and 1995, and then stabilization after that period.

Forestry

According to the National Institute for Forest and Conservation and Development (ICF 2014), the country's long-standing forests cover 48 percent of the land surface (54,000 km²), distributed in four forest macroecosystems: pine forest, broadleaf forest, cloud forest, and mangrove forest. Pine forests (36.3 percent) are mainly found in the western and central highlands, but there are some dense stands at the eastern tip of the country. Broadleaf forests (57.5 percent) are concentrated in the eastern part of the country, but there are small areas throughout the rest of the country. Mixed forests (5.3 percent) and some remnants of cloud forests are found in the center, while the mangrove areas are in the coastal areas in the South (Gulf of Fonseca). The country's north shore covers less than 1.0 percent of its total forest area.

There are three types of property rights over forests: public, municipal, and private. The national forests are largely occupied by people who have property rights but do not have legal title to the property. The municipal forests are often neglected by the municipalities, although there is relatively better control over spontaneous settlement than in the national forests. In private forests, most owners do not process the wood personally; instead, they sell it as standing timber or logs, which brings them very little profit (for example, Jones 2003).

For 2013, ICF (2014) reported that 488,000 cubic meters of logs had been taken from the old-growth forests (97.7 percent). The main logging areas are in the departments of Olancho (25.5 percent) and Francisco Morazán (23.7 percent). An estimated 80,000–100,000 ha of forest are lost annually to the expansion of agriculture, forest fires, and illegal logging, especially in broadleaf forests (Paaby, Hansen, and Florez 2009). Because of replanting, the net deforestation of pine forests is relatively low, but they have suffered a reduction in productivity and genetic quality, mainly as a result of fires, disease, and inadequate selective logging (Paaby Hansen and Florez, 2009).

Historically, the forestry subsector has contributed significantly to the country's GDP through the production and export of whole logs (mostly pine), sawn wood, and resin and its derivatives. In 2000, the forestry sector contributed 1.4 percent to the GDP (\$54 million in exports), representing 5.0 percent of the total agricultural contribution to the GDP (Santos, Gettkant, and Lazo 2007). By 2013, the sector's economic contribution declined to 2.5 percent of the GDP, reflecting the nation's limited attempt and capacity to manage its forests efficiently (Hernández, Velásquez, and Villatoro 2014).

Valuation of agricultural production

Another metric for comparing the importance of agricultural commodities is the value of production (Table 6). One of the difficulties of using this metric is valuing products properly, given that some of the product is consumed by the producers rather than sold on the market. The other difficulty is not adjusting for the cost of inputs—that is, not counting value-added by simply recording gross value. On the other

hand, this metric lets us consider livestock products that were not considered in the harvested area metric.

Figure 21 shows real trends in value of production for the top-five nonlivestock agricultural products, and Figure 22 shows real trends for the top-five livestock products. In Figure 21, we see that maize has remained stable in the value of production, coffee has increased since 1990, and sugarcane has experienced a 150 percent increase since 1990. The gross value of bean production has been irregular throughout the decade, and a trend is not identifiable. In Figure 22, shows a large increase in the value of chicken meat, with almost a sixfold increase from 1990 to 2012. While not as dramatic, the value of milk production more than doubled during the same period, and the growth of beef and eggs was more modest.

Another metric used to rank agricultural products is export value. Table 7 shows the leading export commodities in Honduras, with coffee having the highest value by far. Nevertheless, bananas and oil palm also play a significant role in the Honduran economy. The majority of key imports are food items (Table 8).

Analysis of Crops with Biophysical Models

Biophysical models (also called crop models) were used in the AgMIP GGCM (Rosenzweig et al. 2013) project to determine the effect of climate change—through temperature, precipitation, and solar radiation—on the yields of certain crops. See Thomas et al. (2018) for more on how these models were used.

The bioeconomic model used in our analysis, IMPACT, essentially gives us two types of important projections. The first is that it estimates changes in global supply and demand for agricultural commodities based on assumptions about changes in GDP, population, and improvements in agricultural technology -- without considering climate change. The second is that it does the same, but with the climate effect included. We have much higher confidence in the estimates by IMPACT of the climate effect for crops which are based on the direct estimates from the biophysical models. This section considers two such crops for Honduras: maize and sugarcane.

Maize

Current distribution of production and yield

Maize is the leading crop in Honduras in terms of land area. MapSPAM datasets (You et al. 2014) tell us that 93 percent is rainfed, with the remainder irrigate. Here in the first two subsections, we will focus on rainfed production. From the first map in Figure 23, which shows how much area in each pixel is used for rainfed maize cultivation, we see that it is grown across Honduras, except for the easternmost part of the country. In the second map in Figure 23 which shows the yield distribution for rainfed maize, we see the distribution of yields generally less than 2 MT per hectare, with some very low yields of around 0.25 MT per hectare.

Biophysical modeling of climate effects

Between 2000 and 2050, climate change is expected to reduce yields of rainfed maize throughout Honduras, with a slightly greater impact in the far western part of the country (Figure 24), with pockets

of modest impacts scattered in the central and eastern portions. Careful tabulation of the pixels for each of the four GCMs used in the analysis, taking into account how much maize is grown in each model, reveals that climate change will decrease yields of rainfed maize by 12 percent. However, there was a large range among the results of the four GCMs, with the IPSL model projecting a decline for rainfed maize of 25 percent and the GFDL model predicting only a 6. percent decline. Referring back to the section on climate models, we can attribute the difference between the two primarily to changes in precipitation, which increase in the GFDL and decrease significantly in the IPSL. Higher temperatures in the IPSL model also contribute to the difference.

The median decline for Honduras was very close to the median for the five Central American countries in this study, which projects a median decline of 14 percent. For the global median, our analysis using the AgMIP GGCM data (Rosenzweig et al. 2013) projects a decline of 7 percent.

Impact of climate controlling for technological improvement and global changes in supply and demand

Figure 25 shows the key results of the IMPACT bioeconomic model, taking into account the climate effects computed in the previous section from the biophysical modeling, as well as changed in agricultural technology and global supply and demand. We see that yields could be expected to rise by 67 percent without climate change, but by only 51 percent with climate change. That is, yields will be 9 percent lower with climate change than without. Across the four GCMs, this ranges from losses of 6 percent to 22 percent.

Without climate change, harvested area projected to increase by 30 percent, while with climate change, focusing on the median projection, it will increase by 27 percent. This is ultimately a small difference of 2 percent, and while there is some variation across climate models, it is relatively small, ranging from no change to a loss of 7 percent.

Production will rise by 117 percent without climate change, but with climate change, the median projection still suggests that production will rise by 93 percent. This represents an 11 percent reduction of what could be possible without climate change. The GCMs project a range of losses from 8 percent for the most optimistic climate model to 28 percent for the most pessimistic model.

Figure 26 shows that the world price of maize is expected to increase by 33 percent without and by 54 percent with climate change. The significant range of estimates across climate models is attributed to a decrease in price of 1 percent to an increase of 24 percent.

The projected net exports is a modeled value. Therefore, the actual net exports for the starting year will differ from what was already observed, since the results reflect yields based on average weather for the climate, rather than yields based on actual weather. The bioeconomic model reveals that Honduras is a net importer of maize, and imports are projected under all scenarios to increase over time. Without climate change, maize imports will likely increase by slightly more than 432,000 MT between 2010 and 2050. However, with climate change, that projection will likely be 50,5000 MT higher (at the median projection), though under the most optimistic climate scenario, it may actually be 30,000 MT lower. Figure 26 presents the year-by-year projections for both the world price of maize and net exports of maize.

Summary of findings

Climate will adversely affect both rainfed and irrigated maize production in Honduras. There is some variability in the projected impacts geographically, with the eastern and some of the southern parts of the country less affected. There is also important variation between climate models, with the most negative model showing almost twice the loss in the biophysical model, though only a 60 percent greater loss in the bioeconomic model.

At the median, the negative impact of climate change will be just more than 12 percent of what the yields could be without climate change. While this will not prevent farmers from continuing to produce maize, it will of course present challenges to farmers who depend on their own production for household food security. Effective interventions might include increasing irrigation when technically feasible or helping farmers learn water-retention methods, since much of the projected yield loss was caused by changes in precipitation. Additional interventions might include either rural infrastructure development or policies that enable farmers distant from markets to obtain fertilizers and improved seeds, which they could use to offset yield declines from climate change.

Sugarcane

Current distribution of production

The MapSPAM datasets (You et al. 2014) tell us that 76 percent of the sugarcane in Honduras is rainfed, with the remaining 24 percent being irrigated. In this first part of analysis dealing with sugarcane production in Honduras, we focus on rainfed production. The first map in Figure 27 shows a slightly higher concentration of rainfed sugarcane in the southern and central parts of the country, with very little grown in the East. The second map in the figure shows that yields for rainfed sugarcane are varied across the country, but average around 61 MT per hectare. Similar analysis for irrigated sugarcane (not pictured) reveal average yields around 113 MT per hectare.

Biophysical modeling of climate effects

Climate change is projected to have large, negative consequences for rainfed sugarcane, and that losses appear relatively uniform across the country (Figure 28). The median yield loss from the biophysical model is 37 percent, with projections across climate models ranging from losses of 23 percent to 47 percent. This is slightly smaller than the median loss for the five Central American countries in our study, which is 44 percent. Globally, rainfed sugarcane yields are projected to decline by 29 percent.

Losses for irrigated sugarcane are only slightly less than for rainfed sugarcane, at around 30 percent. This suggests that declines in sugarcane yield are likely to be primarily from increases in temperature, rather than from changes in rainfall.⁵

⁵ In balance, the sugarcane results from the AgMIP Global Gridded Crop Model Intercomparison are based on a single crop model, the Environmental Policy Integrated Climate (EPIC) model. Therefore, they are somewhat less robust than the findings for maize, which are based on four different crop models. Nonetheless, EPIC is a respected model, and the results should not be discounted.

Impact of climate controlling for technological improvement and global changes in supply and demand

Figure 29 presents some of the main findings from the IMPACT bioeconomic model regarding sugarcane in Honduras. It shows a 31 percent improvement in yield with no climate change and a 6 percent decline in yield for the median climate change scenario. That is, adjusting for yield responses to higher prices, and taking into consideration that some of the sugarcane in Honduras is irrigated, the median yield impact of climate change on sugarcane is a loss of 31 percent. IMPACT projects that across the four climate models, losses could range from a high of 36 percent to a low of 16 percent.

Harvested area increases by 65 percent with no climate change and doubles under the median climate change scenario. Growth in area without climate change is largely in response to the rise in global demand and Honduras's ability to produce at high yields making the activity a lucrative use of farmland. Additional increases in area under climate change come about not because Honduras's sugarcane production will not be severely affected by climate, but because global production is also projected to be harmed by climate change, driving up prices, which encourages expansion of land devoted to sugarcane.

These large changes in yield and harvested area lead to even larger changes for production, which grows by 117 percent without climate change and by 79 percent under the median climate change scenario. The maximum climate change scenario leads to just 5 percent less than in the no climate change scenario.

In Figure 30, the price of sugar is expected to rise by 33 percent with no climate change and by 52 percent under the median climate change scenario. Exports of sugar are expected to nearly triple without climate change, but only to rise by 70 percent under the median climate change scenario.

Summary of findings

The productivity of Honduran sugarcane is expected to suffer under climate change, with the direct effects leading to yield declines of around 37 percent for rainfed production (slightly higher than the 29 percent global average, and lower than some of its neighbors). The reason for the differences between Honduras and the rest of the world in climate impacts on sugarcane productivity is a complex combination of the impact of the change in both rainfall and temperature over the growing cycle of sugarcane – differences that are subtle and can only be found through the use of complex crop models with daily time steps, like those used in the AgMIP GGCMI (Rosenzweig et al. 2013). But the main reason for reduction in yields in Honduras as well as the rest of the world is a rise in temperatures.

In principle, breeders, if given enough time, should be able to develop varieties that are less sensitive to temperature. This is something that investment – both public and private and at a global and regional scale – could accomplish. The role of the government is to ensure that Honduras stays up to date on new varieties and production techniques that may be developed, and that they facilitate the promulgation of new varieties and techniques to farmers.

While the IMPACT bioeconomic model, taking into consideration gains and losses in other crops, as well as changes in global prices, projects that Honduran farmers will optimally choose to dramatically expand their cultivated area of sugarcane, there appears to be quite a large amount of uncertainty surrounding the future of sugarcane under climate change. With very large projected declines in sugarcane productivity in Honduras, if, for example, another region of the world would develop a comparative advantage as a result of developing new varieties that were only suited to that region (or with proprietary

knowledge not shared with other regions), then it might not be optimal for Honduras to expand land that it would be at a comparative disadvantage in. That is, it would not be wise for the government to seek to establish policies that would unduly encourage the expansion of sugarcane area until sugarcane productivity effects of climate change are first dealt with.

Analysis of Crops Lacking Biophysical Models

For the crops presented in this section, we did not have biophysical models available. So to proxy some climate effect, we used a composite of the results of climate impacts on other crops for which we did have biophysical models. In particular, for palm oil and coffee, the climate effect used in the bioeconomic model is the area weighted average of barley, cassava, groundnuts, rice, soybeans, and wheat. For beans, we used the area weighted average of soybeans and groundnuts.

Because this methodology may not perform very well, when possible, we also present research from other studies that attempted to model these crops. We believe that economic analysis for the scenario that ignores climate change to be reliable, and recommend using with some caution the climate change effects for these crops that are not based on their own biophysical models.

Coffee

Current distribution of production

In the first map in Figure 31, rainfed coffee is grown throughout Honduras, with a somewhat lower concentration in the easternmost part of the country. The second map in Figure 31 shows that yields vary throughout the country, but on average they are just over 800 kilograms per hectare.

Modeling changes in the supply of and demand for coffee

Because we were unable to model the direct biophysical impacts of climate change on coffee, they were inferred from the impacts of climate change on other crops. Thus, rather than drawing strong conclusions about the impacts of climate change on coffee, we focus more on the global market effects in the coffee sector, which is reported as the “no climate change” scenario. Nevertheless, we mention the climate effects modeled in IMPACT, as we can at least consider what those effects might be, even if our confidence in these results is not as high.

Figure 32 shows that coffee yields increase slightly under the no climate change scenario, by only 5 percent. Under the median climate change scenario, they are only projected to be 1 percent higher, with a very narrow range on the climate projections, from no increase over the no climate change scenario to a 2 percent increase in the best case scenario. Harvested area is expected to increase modestly by 6.6 percent without climate change and by 8 percent with climate change. Therefore, coffee production is expected to rise by 12 percent under the no climate change and 14 percent for the median climate change scenarios.

The price of coffee is expected to rise by 32 percent with no climate change and by 42 percent under the median climate change scenario (Figure 33). Exports are expected to decline by slightly more than 5 percent between 2010 and 2050 without climate change, and by only 2 percent under the median climate change scenario.

Summary of findings

Within IMPACT and not taking into account climate change, both yield and harvested area are expected to increase between 2010 and 2050, but only modestly, leading to a slightly larger increase in coffee production. Given the 32 percent price increase that is projected over the same period, there are economic incentives to develop policies or investments that might improve Honduras' ability to increase its production.

However, Ovalle-Rivera et al. (2015) suggest that by 2050, climate change will reduce the median area suitable for growing coffee in Honduras by 26 percent (if we make the major assumption that lost area is equivalent to yield reduction resulting from climate change), and is among the larger losses reported in their analysis—though they project every coffee-producing country will lose suitable area. This is clearly a much larger value than was reported in IMPACT. The analysis of Ovalle-Rivera et al. (2015) covered enough coffee-growing nations to account for 89 percent of the world's coffee. If the reduction of suitable area roughly corresponds to the impact of climate on coffee productivity, then globally (using the countries in their study), we would expect that the direct impact of climate change on coffee will be to reduce yields by 21 percent.

As a result, IMPACT underestimates the true price effect of climate change, and underestimates the negative shock to Honduras' agriculture sector. Ovalle-Rivera et al. (2015) tell us that it will be necessary for Honduran coffee growers to move to higher elevations, which they remind us are often forested, and therefore could result in conversion of long-standing forests to agricultural land.

Läderach et al. (2013, p. 1) conclude that "unless additional efforts are made to strengthen adaptive capacity today, there will likely be heavy economic losses across the coffee supply chain, as well as the disappearance of important ecosystem services.... Given the long lead time for investments, actions must be taken now." Their policy recommendations, which we endorse, include:

- Developing climate stress-resistant coffee varieties, validating agronomic management strategies, and improving market links.
- Providing financial assistance via subsidies, insurance, and ecosystem payments (through either direct remuneration, or the development of markets to reward sustainable land practices and forest conservation).
- Promoting diversification as a short-term risk management strategy and a long-term bridge to full crop substitution.

Beans

Current distribution of production

The first map in Figure 34 shows that rainfed beans are concentrated in the western and southern parts of the country. The second map shows that yields are variable across the country, but tend to be lower in places that have low concentrations of bean production. Yields average around 750 kilograms per hectare for the country, at least according to data calculations based on the two maps (authors, from You et al. 2014).

Impact of climate controlling for technological improvement and global changes in supply and demand

Beans were not modeled directly in the biophysical models. Unlike for the case of coffee, beans have a much closer similarity to other legumes than does coffee to annual crops. Therefore, while we still hold a degree of skepticism about the effects of climate on beans used in the IMPACT model -- since they were not obtained directly from biophysical models of beans -- we nonetheless have an expectation that the estimates have a reasonable connection to biophysical model results. Thus, we should not disregard what the IMPACT bioeconomic model tells us about the climate effects.

Figure 35 shows a 55 percent increase in yield with no climate change, but only a 39 percent increase in the median climate change scenario. This means that the best guess is that climate change will lead to 10 percent lower bean yields, though across climate models, this estimate ranges from 8 percent to 20 percent lower. Harvested area increases by 47 percent without climate change and by 44 percent with climate change at the median, while production increases by 126 percent without climate change and by 100 percent with climate change at the median—or 12 percent lower than without climate change. However, the effects of climate on production range from 8 percent lower to 26 percent lower.

Figure 36 shows changes in the world price for beans. The price of beans with no climate change increases by 14 percent, while it increases by 32 percent in the median climate change scenario. The trade deficit is expected to improve without climate change by around 15 MT, to worsen by 4 MT with the median value under climate change, but to worsen by 33 MT under one climate model.

Summary of findings

The effects of climate on beans in Honduras are projected under most climate models to be only modestly negative. Given that global prices are also projected to rise only modestly, it appears that little intervention is needed in the bean sector. However, it is important to keep in mind that IMPACT includes assumptions about growth in bean productivity through time, because it assumes that some additional research will be conducted to support growth. Therefore, research on improving bean yields should not be completely neglected, or the model could prove to be overly optimistic about future bean yields in Honduras.

Palm Oil

Current distribution of production

In Figure 37, the first map shows the distribution of palm oil. We note that it is concentrated in the central part of Honduras. The second map in the figure shows that yields throughout the country appear to be generally uniform. Averaging over the entire country, the data tells us that mean yield is 15 MT per hectare.

Modeling changes in the supply of and demand for palm oil

As with the coffee analysis, the effects of climate on palm oil yield are based not on direct modeling by a biophysical model, but on imputed results from other crops. Therefore, we should be cautious in drawing any strong inferences concerning the results. Instead, we focus on the no climate change scenario, which is based on more robust estimates concerning changes in palm oil supply and demand.

Without climate change between 2010 and 2050, palm oil yield is expected to increase by 19 percent, almost the same as projected for the median climate change scenario; harvested area is expected to increase by 52 percent, with no climate change effect; and production is expected to increase by 81 percent, with no appreciable impact of climate change (Figure 38). Also important to note is that production is projected to rise fairly rapidly between 2010 and 2030, and then much more slowly from 2030 to 2050. This reflects the patterns observed in yields and harvested area growth, though the shift in the speed of growth occurs around 2025 for harvested area.

The world price for palm oil is expected to increase by roughly 30 percent under the no climate change scenario, and the climate scenarios have almost identical price trends (Figure 39). As noted for harvested area trends in Honduras, the fastest period of growth in palm oil price is between 2010 and 2025, with a more gradual rise thereafter.

Net exports of palm oil are expected to rise by around 67 percent until 2030–2035, and then decline through 2050, with the 2050 level with no climate change roughly 20 percent higher than the 2010 level of net exports (Figure 39). The median climate scenario suggests that exports would not be much different than without climate change, while the full range of climate impact on net exports suggest that they could be anywhere from 14 percent to 40 percent higher than the 2010 value.

Summary of findings

Palm oil production will likely expand in Honduras, through both increased yields and increased cultivated area. Change will be most rapid through 2030 (due, at least in part, to the degree of change in world prices), and will be more gradual thereafter. Our not assuming a strong impact of climate change on productivity was reflected in the modest differences with the no climate change scenario. Based on this analysis, we have no policy recommendations. However, if any kind of major investment of public funds in expanding oil palm were to be made, it would be important to use other means to analyze the potential impacts of climate on palm oil yield. One possibility would be to conduct statistical analyses of palm oil productivity using historical data across countries and regions as a function of temperature and precipitation.

Livestock

Cattle

Current distribution of production

The highest concentrations of grasslands in Honduras are in the eastern and in the west central areas (Figure 40). However, the highest concentrations of cattle are in the South, though they appear in substantial numbers throughout the country, except in the East (Figure 41).

The effect of climate on the median yield of rainfed managed grasses is a sort of proxy for how climate might affect the productivity of grazing cattle (Figure 42). In general, pastures will not be largely affected by climate change. In the East, where cattle concentrations are low, grassland productivity is projected to increase. There is also some productivity growth in the southern part of the country.

Modeling changes in the supply of and demand for beef

The IMPACT bioeconomic model does not model the direct effects of climate on cattle yet. Therefore, most of the effects attributed to climate change actually reflect its impact on livestock feed prices.

Between 2010 and 2050, beef production in Honduras is projected to increase by 119 percent (Figure 43). Beef prices globally are projected to grow by 20 percent during the period, peak during 2035–2040, and then decline through 2050 to 15 percent above the 2010 price.

Beef imports are expected to rise through the late 2030s, and then the trend reverses, with Honduras becoming a net exporter by 2050. Trade volume, however, is projected to be modest, with net imports peaking at 5,000 MT, and net exports peaking at slightly less than 3,000 MT.

Modeling changes in the supply of and demand for milk

Between 2010 and 2050, milk production is projected to grow at a slower rate than beef production, at around 39 percent (Figure 44). However, similar to global pricing for beef, milk prices will peak around 2040, at about 14 percent higher than in 2010, and will decline slightly through 2050, to 12 percent higher than in 2010. Imports are projected to rise, almost tripling from 2010 to 2040, and then stabilize or fall slightly through 2050.

Summary of findings

Without the full effects of climate on cattle productivity taken into consideration, we see that IMPACT projects a steady rise in beef production, more than doubling 2010 production by 2050. Milk production is projected to expand more modestly. We do not have separate data on the locations of beef versus dairy operations, so the following comments apply to both, except that dairy cows are generally more sensitive to heat stress than beef cattle. Heat stress not only lowers the output of milk, but also reduces its fat and protein content (Key, Sneeringer, and Marquardt 2014; Hatfield 2008; Kadzere et al. 2002; St-Pierre, Cobanov, and Schnitkey 2003; West 2003).

Because IMPACT did not model direct climate effects on livestock, we can use some rules of thumb in reasoning about what they might be. Generally, we know that as temperatures in tropical and subtropical areas rise, livestock production will be hindered in three main ways. First, higher temperatures can stress the animals, leading to lower meat and milk productivity and lower reproduction rates (Key, Sneeringer, and Marquardt 2014; Chase 2006), as well as higher mortality rates (Crescio *et al.* 2010). Second, higher temperatures increase susceptibility to disease through weakened immune systems (Climate Change Connection 2013; Lemmen et al. 2008). And third, higher temperatures can increase the prevalence and range of livestock diseases (Key, Sneeringer, and Marquardt 2014). We are not certain how to apply the second and third points to the case of Honduras, since we are unable to project the climate impact on parasite and pathogen prevalence; therefore, we will focus mostly on the first point.

Referring back to the weather and climate change maps presented earlier in this report (Figures 10–15), we note that in the southern part of the country, mean daily maximum temperature in the hottest month of the year already ranges in the 36–37 °C range. With an additional 2–3 °C by 2050, heat stress may reduce the productivity of cattle in that region.

As such, it may be optimal to shift cattle from the warmer parts of the country into the cooler parts, at least during the hottest months of the year. Key, Sneeringer, and Marquardt (2014) suggest that several

other strategies can help cows adapt to higher temperatures and offset productivity losses from heat. For example, farmers could invest in structures or plant trees to provide shade, though trees may be preferred because they provide better air circulation and can often have better cooling effects because of evapotranspiration. They could also invest in fans or cooling sprays (Flamenbaum et al. 1995; Her et al. 1998; Igono et al. 1987).

Chickens

Current distribution of production

Figure 45 shows that chicken production is dispersed evenly throughout the country.

Modeling changes in the supply of and demand for meat from chickens

As noted for cattle, IMPACT does not model the direct effects of climate on chickens. Thus, we focus not on the climate effect, but on the general effects of changes in productivity and changes in global supply and demand. IMPACT projects that production of chicken meat will rise by 149 percent between 2010 and 2050 (Figure 46). This rise is at least partly in response to the rise in global prices for poultry, which are projected to increase by 24 percent over the same period.

Imports of chicken meat are projected to rise by 163 percent between 2010 and the late 2030s, and then fall suddenly, to the extent that Honduras could become a net exporter by 2050. Volumes of imports and exports are modest, however.

Modeling changes in the supply of and demand for eggs from chickens

Figure 47 shows similar changes in production, prices, and net exports of chicken eggs as for poultry, though they are far from identical. For example, egg production is projected to rise more modestly than poultry meat production, with egg production rising by 89 percent between 2010 and 2050.

Global prices for eggs peak around 2035 at only 12 percent higher than in 2010, and then taper slightly through 2050 to around 6 percent higher than in 2010. Imports rise through the late 2020s, then level off until around 2040, after which they decline to 2050, settling at almost the identical level of 2010. Import volume is relatively modest for eggs.

Summary of findings

Chicken production will become an increasingly important part of the agriculture sector in Honduras, with significant growth in the production of both poultry meat and eggs. As with cattle, chickens are sensitive to heat stress, which can slow weight gain in broilers (Key, Sneeringer, and Marquardt 2014; Cooper and Washburn 1998; Yalcin et al. 2001; Quinteiro-Filho et al. 2010; Sohail et al. 2012); reduce the weight of eggs in layers (Key, Sneeringer, and Marquardt 2014; Bogin et al. 1996); and increase the mortality rate of the chickens (Key, Sneeringer, and Marquardt 2014; Bogin et al. 1996).

Climate change may cause a shift in where chickens are raised, especially for commercial production, from areas with higher temperatures to areas with more moderate temperatures. Investments in cooling will help in adapting chicken production to climate change. It may also be possible to develop or import new varieties of chickens that are not as sensitive to heat. Finally, reducing the density of chickens in barns would cool the barns, as fewer birds would emit less heat (Key, Sneeringer, and Marquardt 2014).

Changes in Food Security

Figure 48 shows projections for both the share of the Honduran population at risk of hunger and the total number of Hondurans at risk. These projections are from IMPACT and are based on Fischer et al. (2005), who use a quadratic formula for kilocalories consumed. Kilocalories, in turn, are based on income and food prices.

The share of the population at risk of hunger declines gradually, at first, and then the rate of decline accelerates after 2020. By 2050, without climate change, the share at risk declines by 59 percent, affecting 8.7 percent of the population. However, with climate change the share at risk is projected to be 9.4 percent of the population at the median value of the projections from the four climate models.

In terms of number of people at risk of hunger, there is a projected rise in the number up to around 2020, and then a decline thereafter. Overall, the absolute number at risk is projected to fall by around 36 percent between 2010 and 2050 without climate change, and by 31 percent with climate change evaluated at the median value.

Overall, the reduction in food insecurity by 2050 is a very positive sign. However, with around 9 percent of the population being food insecure—slightly more than a million people—the numbers are sufficiently large to challenge policy makers to do even more. Climate change is projected to increase food insecurity in three of the four models, but the median increase is relatively modest.

Conclusions

Our analysis of the future impact of climate change on crops and livestock has endeavored to quantify the likely effects of climate change on important components to the agricultural sector so that policy makers and donors might be able to prioritize action steps. Clearly climate will lower yields in Honduras below what could otherwise be obtained without climate change, but climate change will have this effect on yields globally. For some key crops, such as maize and sugarcane, yield reductions will likely be larger in Honduras than on a global level. Nonetheless, the models suggest that the degree of yield reduction in Honduras is not high enough that local production will be curtailed. They do suggest, however, that policymakers need to have an active role in using policies and investments to best help farmers.

We expect that the two highest-value crops for Honduras—coffee and sugarcane—may also be the hardest hit by climate change. The main recommendation for sugar cane is for heat resistant varieties to be developed. And for coffee, in the absence of new varieties, it will generally need to be grown at higher altitudes than it is currently grown. This means that farmers who grow coffee at lower altitudes will no longer be able to grow coffee. One possibility is for the government to assist coffee farmers to acquire land at higher elevations, if they want to relocate. Another option is to help farmers find alternatives to growing coffee at lower elevations, which will require interacting with them to learn what alternatives they would like to pursue. Possibilities include raising livestock, planting annual food crops, or finding land suitable for an alternative tree crop.

Maize is projected to have a productivity loss of around 12 percent as a direct result of climate change. Since so many farmers, particularly subsistence farmers, depend on maize for their livelihood, it is critical that investment be made in developing heat-resistant varieties that will remain productive despite changes in climate. Indeed, investment in research is critical if the negative effects of climate change are

to be minimized. And for some crops – especially those with high commercial value – creating laws that facilitate private enterprise developing new varieties on their own is of paramount importance

Very little irrigation is currently available in Honduras. As economically and hydrologically feasible, investment in irrigation could serve to intensify production and help farmers manage risk in drier years. Three out of the four climate models suggest that Honduras will get drier with climate change. On the other hand, it is important to keep in mind that the main challenge Honduran farmers will face in the future is higher temperatures and not too little rainfall.

Special care needs to be taken with regard to livestock production in certain parts of Honduras, particularly the South, where heat stress could easily reduce the potential for animal production. Alternative breeds or alternative husbandry methods that create cooler environments at least during the hottest part of the day in the hottest part of the year would be very important.

All of the recommendations are undergirded with the assumption that there will be effective means of communicating new ideas and new methodologies to farmers. This can be done through traditional agricultural information services, but many of these ideas might be transferred through the use of radio, television, and mobile phones.

Finally, it is important to understand that policies that serve in some way to enhance productivity in agriculture are policies that are potentially invaluable in helping farmers compensate for the negative effects of climate change. These might include road construction and maintenance, but they could also include coordinating access to international markets for some commodities, and lowering transaction costs for credit and inputs. Better connections of farmers to markets lowers input prices at the farmgate, raises output prices at the farmgate, and helps make use of better inputs more economically viable.

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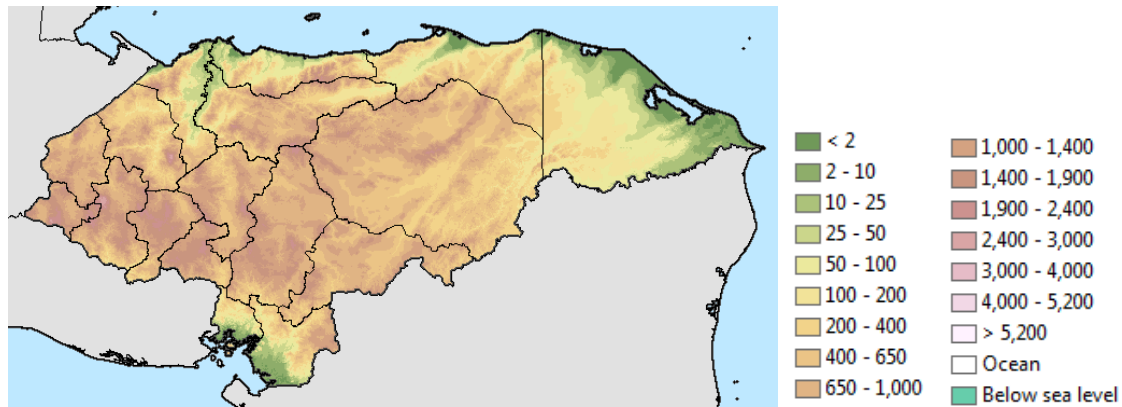
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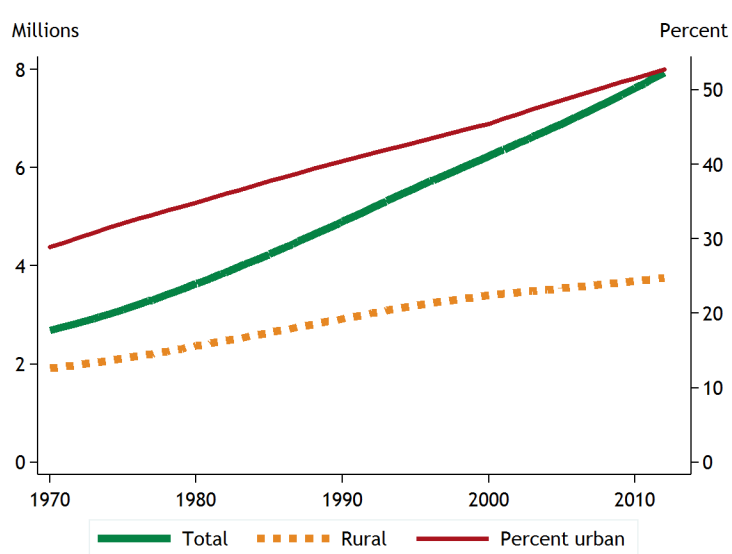
FIGURES

Figure 1 Elevation of Honduras (meters)



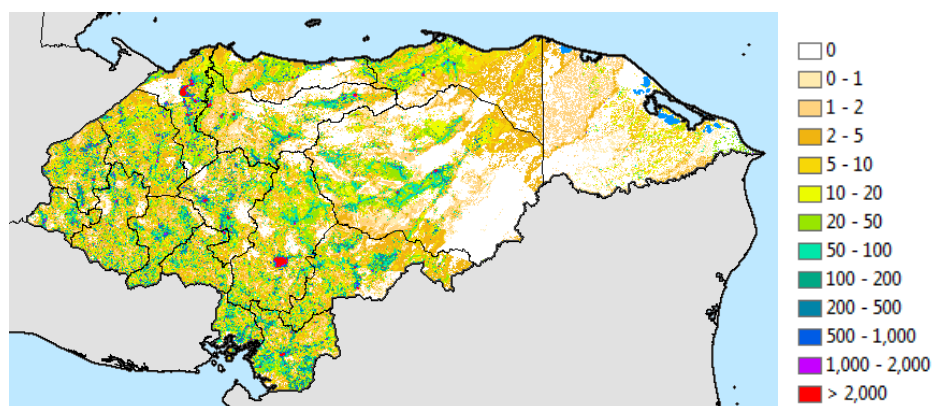
Source: GLOBE Task Team et al. (1999).

Figure 2 Population trends in Honduras: total and rural populations (millions of people) and urban population (percent), 1970–2012



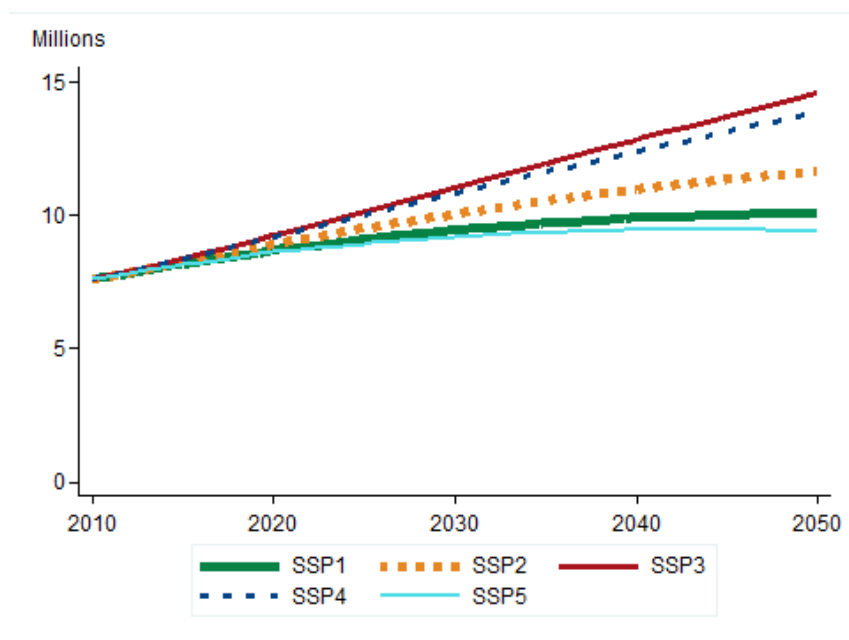
Source: World Development Indicators (World Bank 2014).

Figure 3 Population distribution in Honduras, 2012 (people per square kilometer)



Source: LandScan (ORNL 2013).

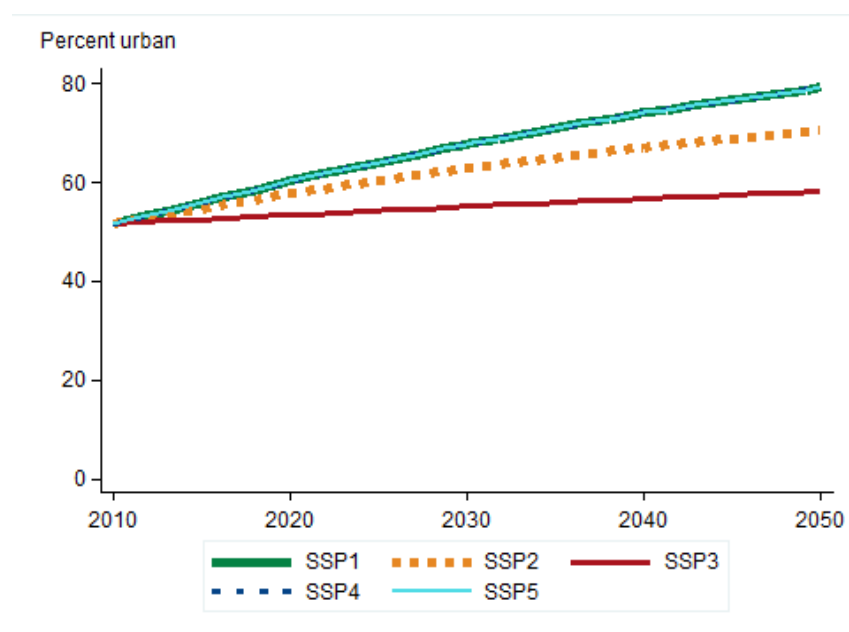
Figure 4 Population projections for Honduras, 2010–2050



Source: IPCC (2014).

Note: SSP = shared socioeconomic pathway.

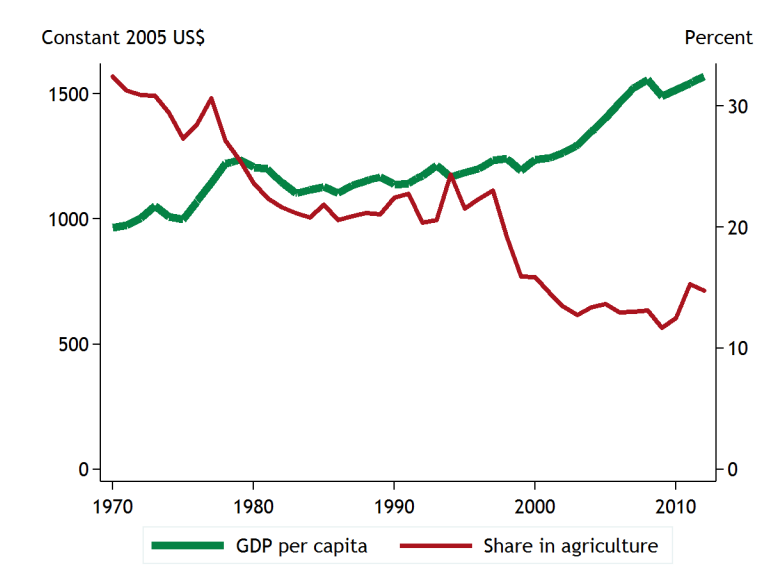
Figure 5 Urbanization projections for Honduras, 2010–2050



Source: IPCC (2014).

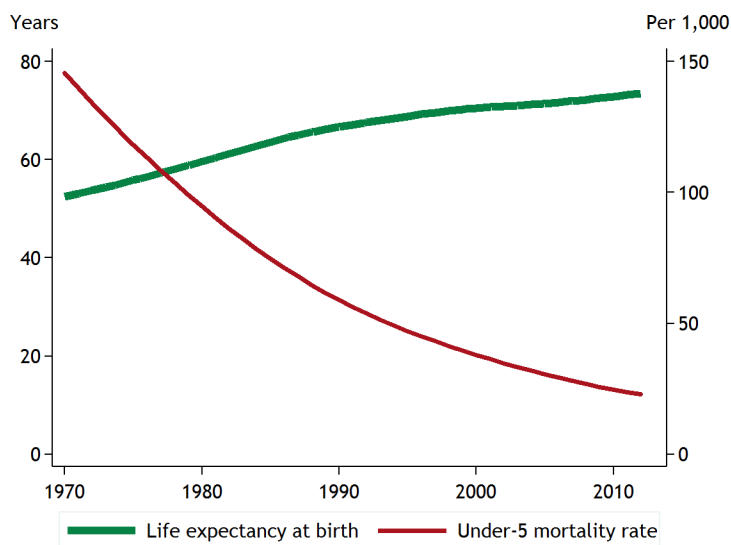
Note: SSP = shared socioeconomic pathway.

Figure 6 Per capita GDP in Honduras (constant 2005 US\$) and share of GDP from agriculture (percent), 1970–2012



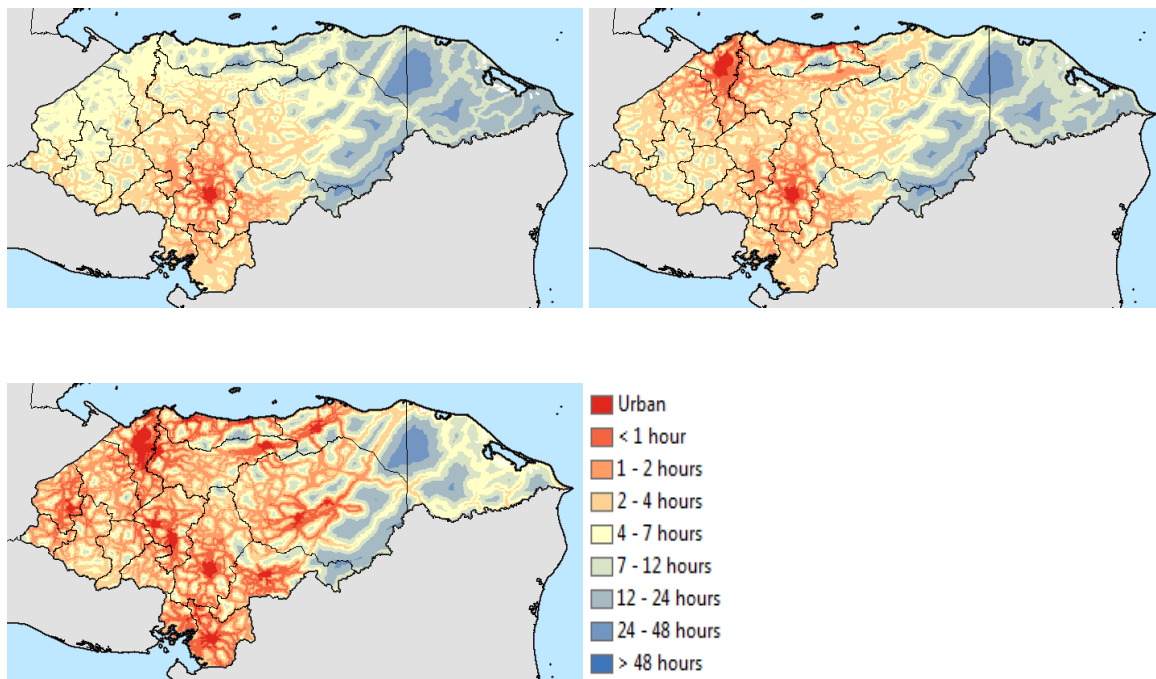
Source: World Development Indicators (World Bank 2014).

Figure 7 Well-being indicators for Honduras, 1970–2012



Source: World Development Indicators (World Bank 2014).

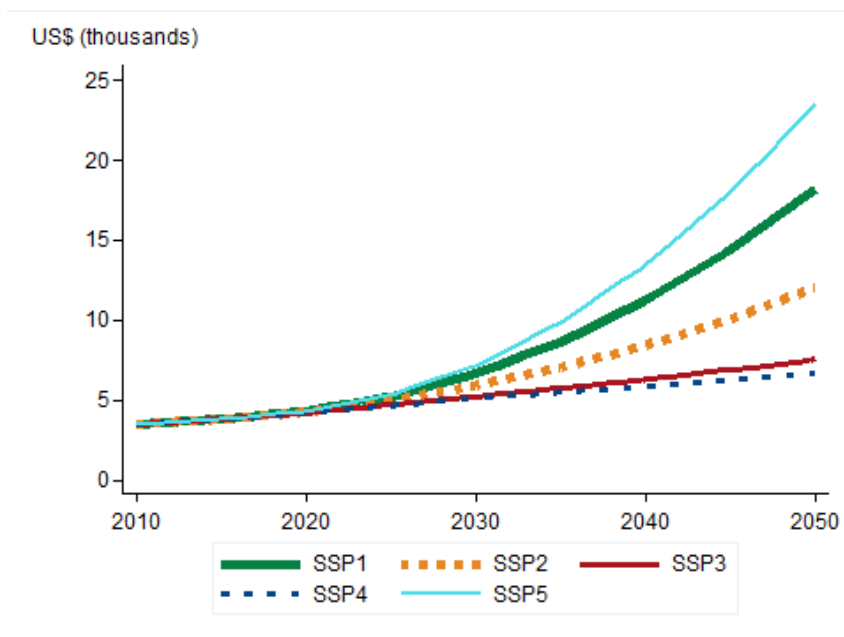
Figure 8 Travel time to urban areas in Honduras circa 2000 (top left graph looked inverted)



Source: Guo and Cox (2014).

Note: Top left is to cities of 500,000 or more people; top right is to cities of 100,000 or more people; bottom left is to towns and cities of 20,000 or more people.

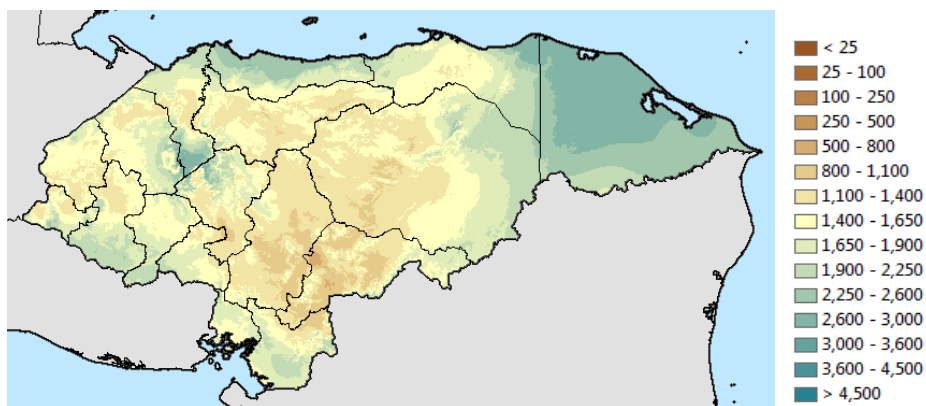
Figure 9 GDP per capita in Honduras, future scenarios, 2010–2050



Source: IPCC (2014).

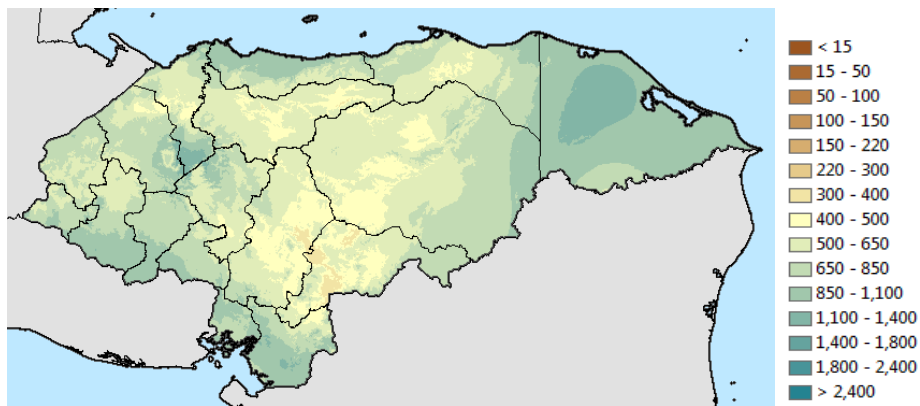
Note: The gross domestic product value is purchasing power parity. SSP = shared socioeconomic pathway.

Figure 10 Annual rainfall (millimeters) in Honduras, 1960–1990



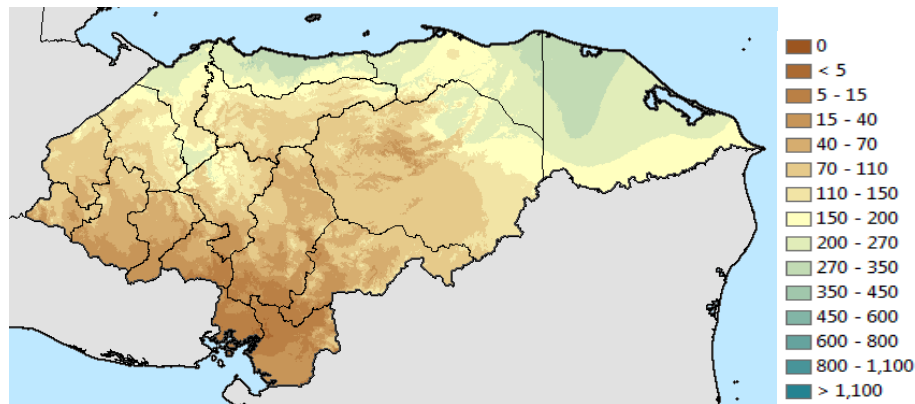
Source: WorldClim version 1.4 (Hijmans et al. 2005).

Figure 11 Rainfall (millimeters) for the wettest three months in Honduras, 1960–1990



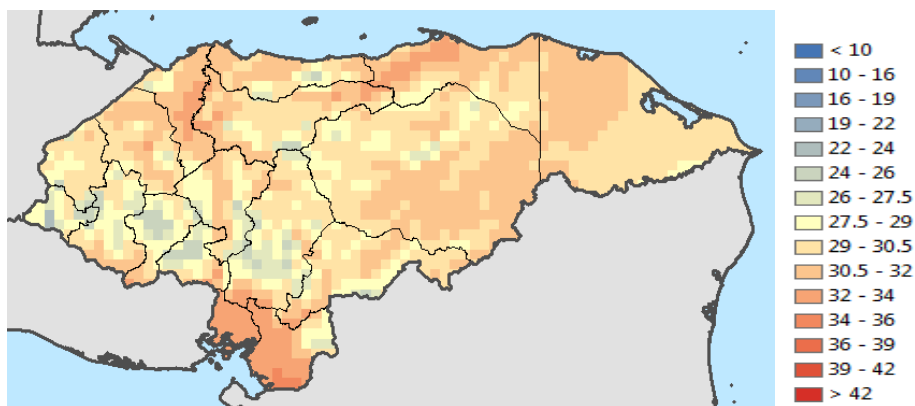
Source: WorldClim version 1.4 (Hijmans et al. 2005).

Figure 12 Rainfall (millimeters) for the driest three months in Honduras, 1960–1990



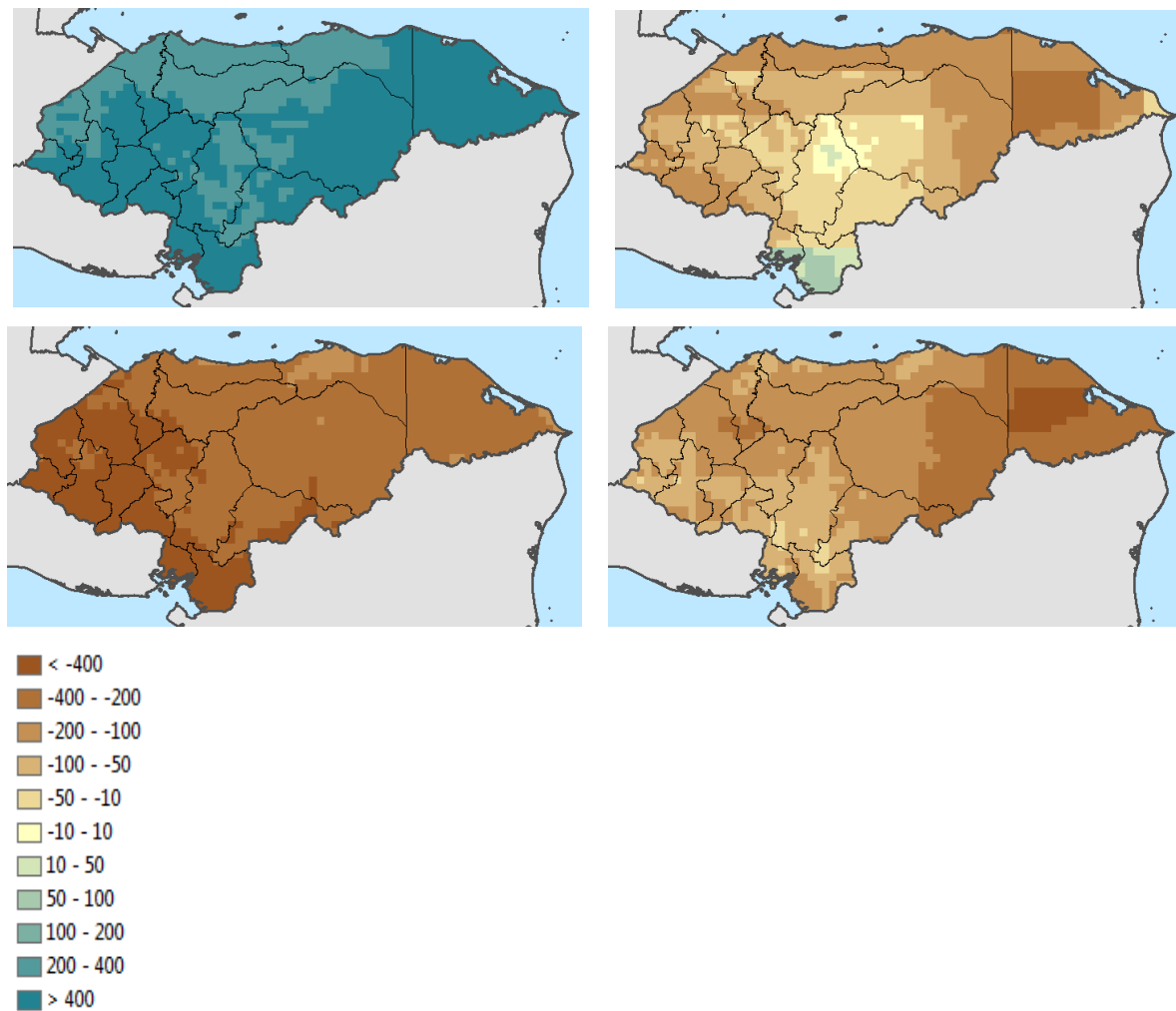
Source: WorldClim version 1.4 (Hijmans et al. 2005).

Figure 13 Mean daily maximum temperature (degrees Celsius) for the warmest month in Honduras, 1960–1990



Source: WorldClim version 1.4 (Hijmans et al. 2005).

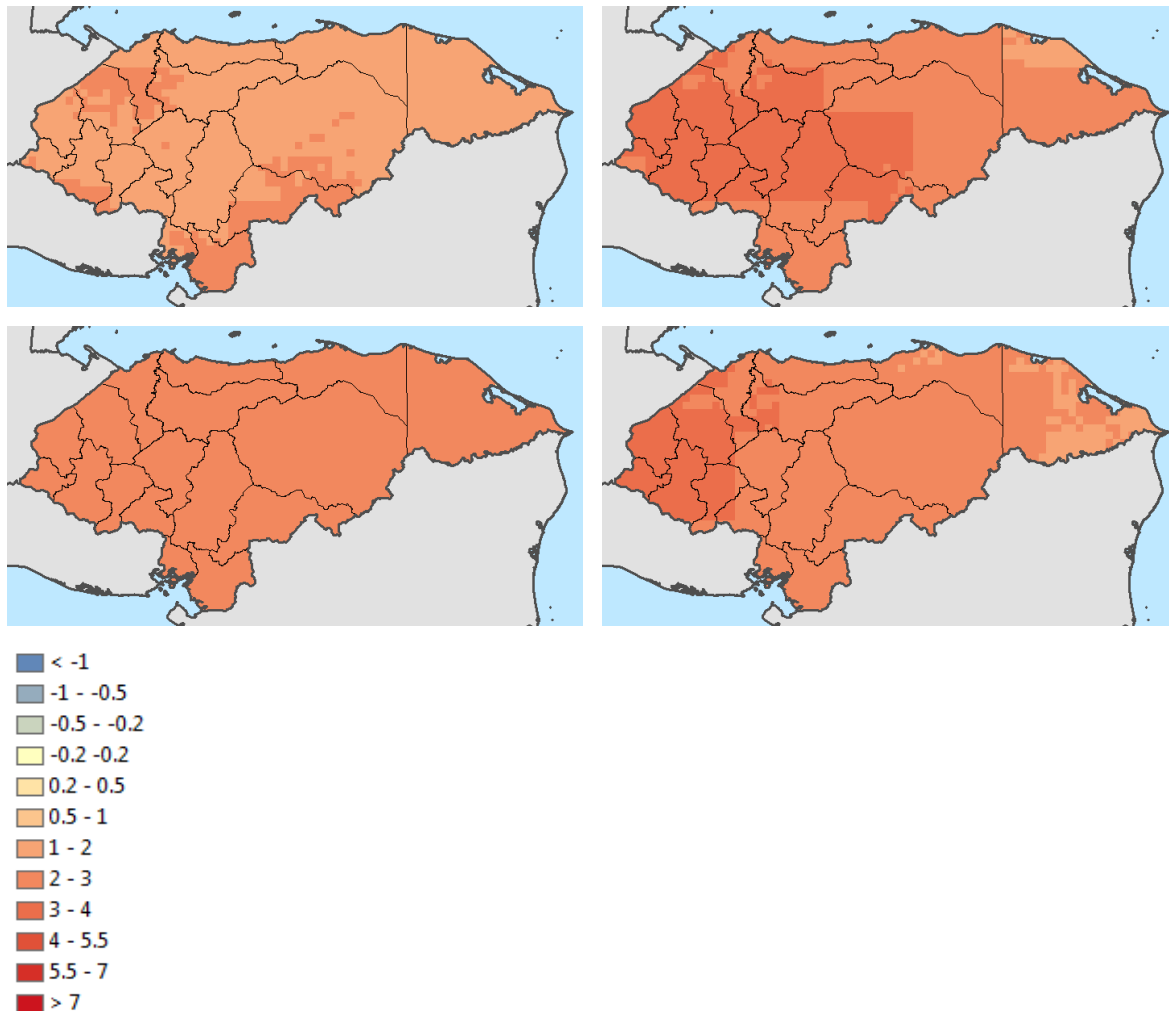
Figure 14 Changes in mean annual precipitation (millimeters) in Honduras, 2000–2050



Source: CMIP5.

Note: Top left, Geophysical Fluid Dynamics Laboratory (GFDL); top right, Hadley Centre Global Environmental Model (HadGEM); bottom left, L'Institut Pierre-Simon Laplace (IPSL); bottom right, Model for Interdisciplinary Research on Climate (MIROC). All are from the Coupled Model Intercomparison Project (CMIP) for the Agricultural Model Intercomparison and Improvement Project (AgMIP), and are for Representative Concentration Pathways, version 8.5 (RCP8.5).

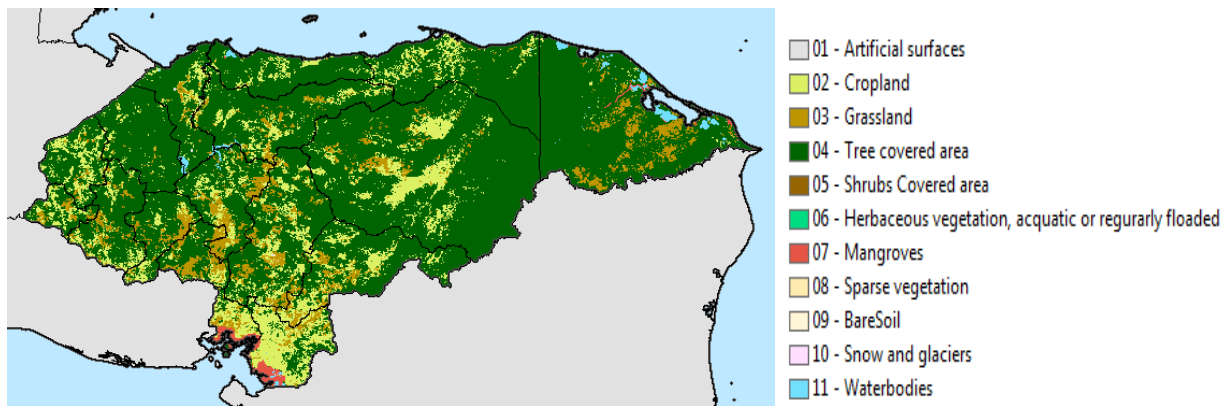
Figure 15 Changes in mean daily maximum temperature for the warmest month in Honduras, 2000–2050



Source: CMIP5.

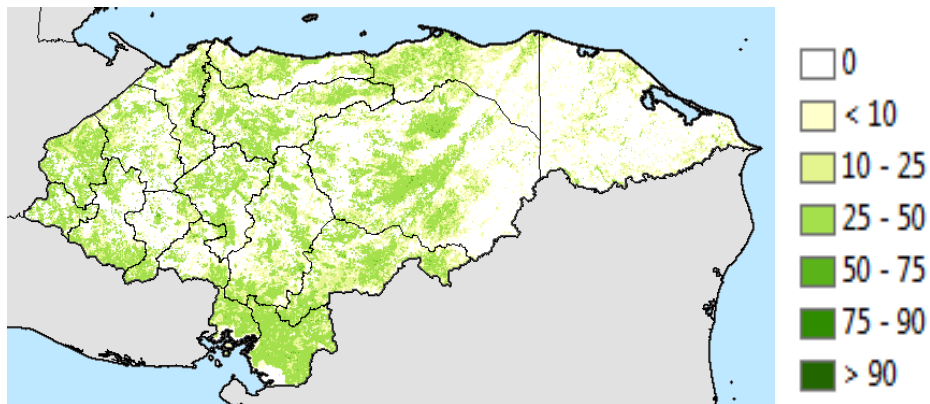
Note: Top left, Geophysical Fluid Dynamics Laboratory (GFDL); top right, Hadley Centre Global Environmental Model (HadGEM); bottom left, L’Institut Pierre-Simon Laplace (IPSL); bottom right, Model for Interdisciplinary Research on Climate (MIROC). All are from the Coupled Model Intercomparison Project (CMIP) for the Agricultural Model Intercomparison and Improvement Project (AgMIP), and are for Representative Concentration Pathways, version 8.5 (RCP8.5).

Figure 16 Dominant land cover in Honduras, 2010



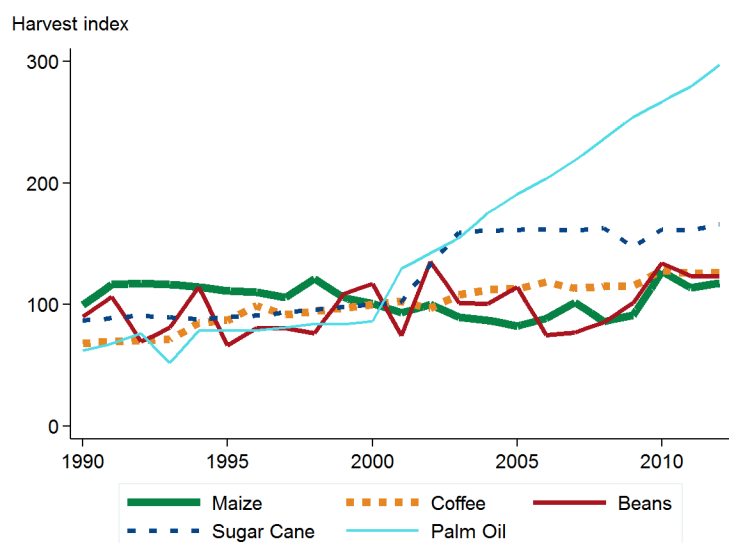
Source: GLC-SHARE (Latham et al. 2014).

Figure 17 Percentage of cropland in Honduras in 2010



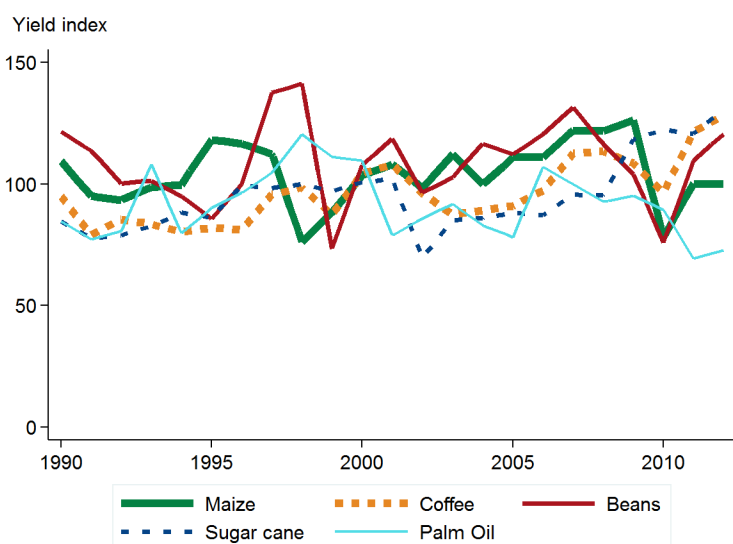
Source: GLC-SHARE (Latham et al. 2014).

Figure 18 Harvested area trends in Honduras, 1990–2012



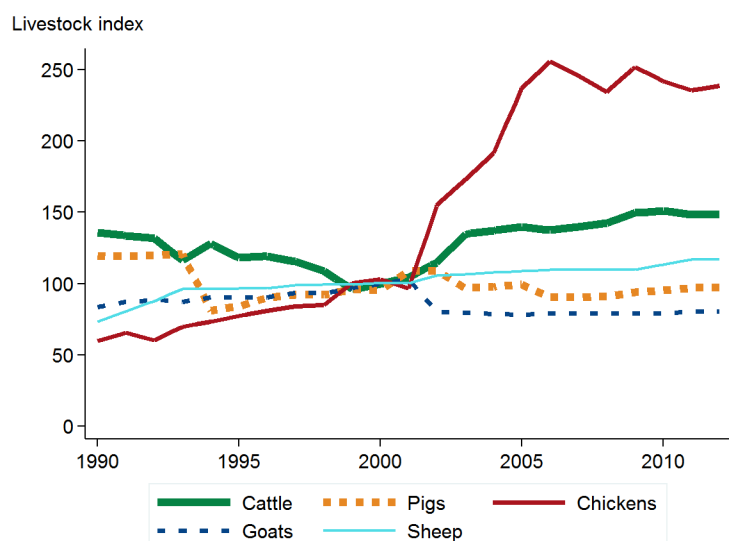
Source: FAOSTAT (FAO 2014).

Figure 19 Yield trends in Honduras, 1990–2012



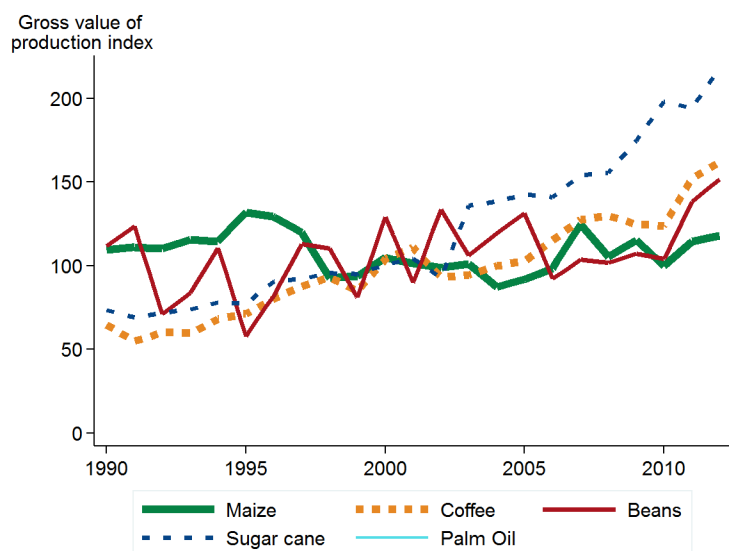
Source: FAOSTAT (FAO 2014).

Figure 20 Livestock trends in Honduras, 1990–2012



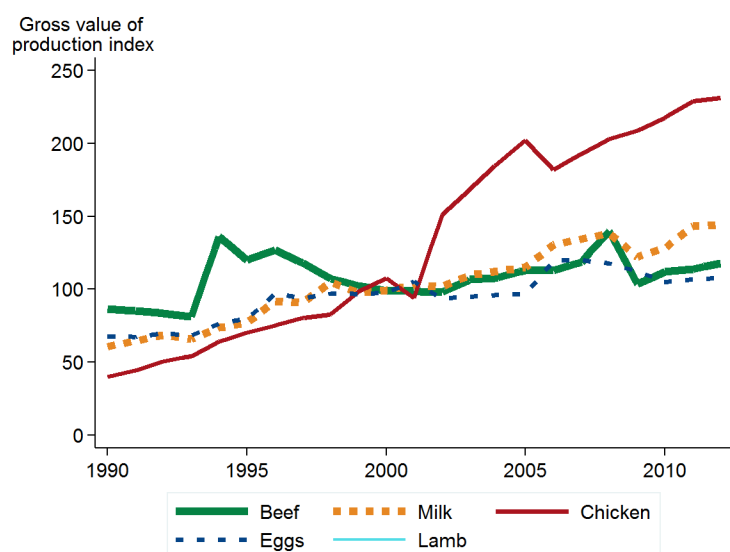
Source: FAOSTAT (FAO 2014).

Figure 21 Trends for gross value of production for highest-value nonlivestock products for Honduras, 1990–2012



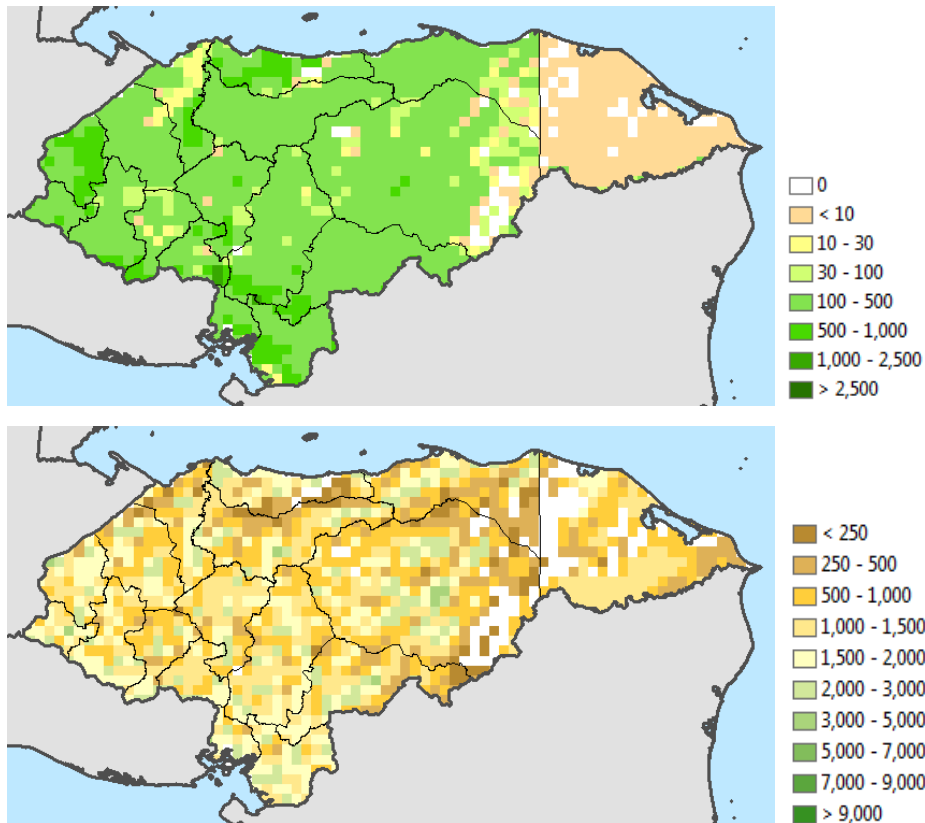
Source: FAOSTAT (FAO 2014).

Figure 22 Gross value of production for livestock in Honduras, 1990–2012



Source: FAOSTAT (FAO 2014).

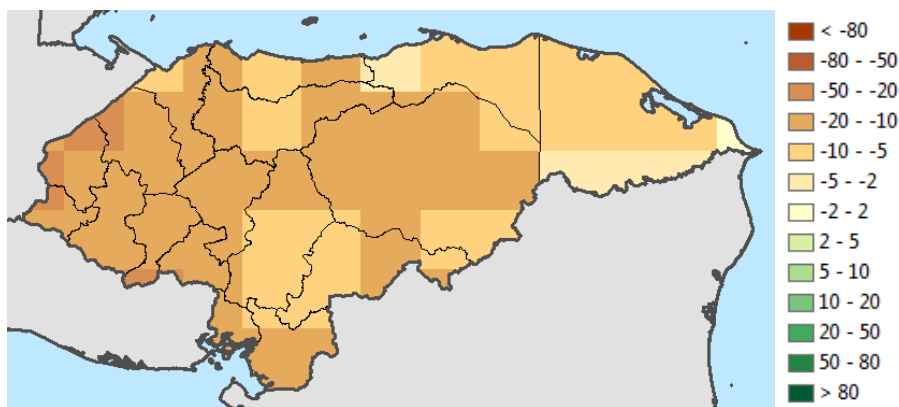
Figure 23 Intensity (hectares per pixel) and productivity (kilograms per hectare) of rainfed maize in Honduras, circa 2005



Source: Spatial Production Allocation Model (SPAM, You et al. 2014).

Note: A pixel at the equator contains approximately 8,500 hectares.

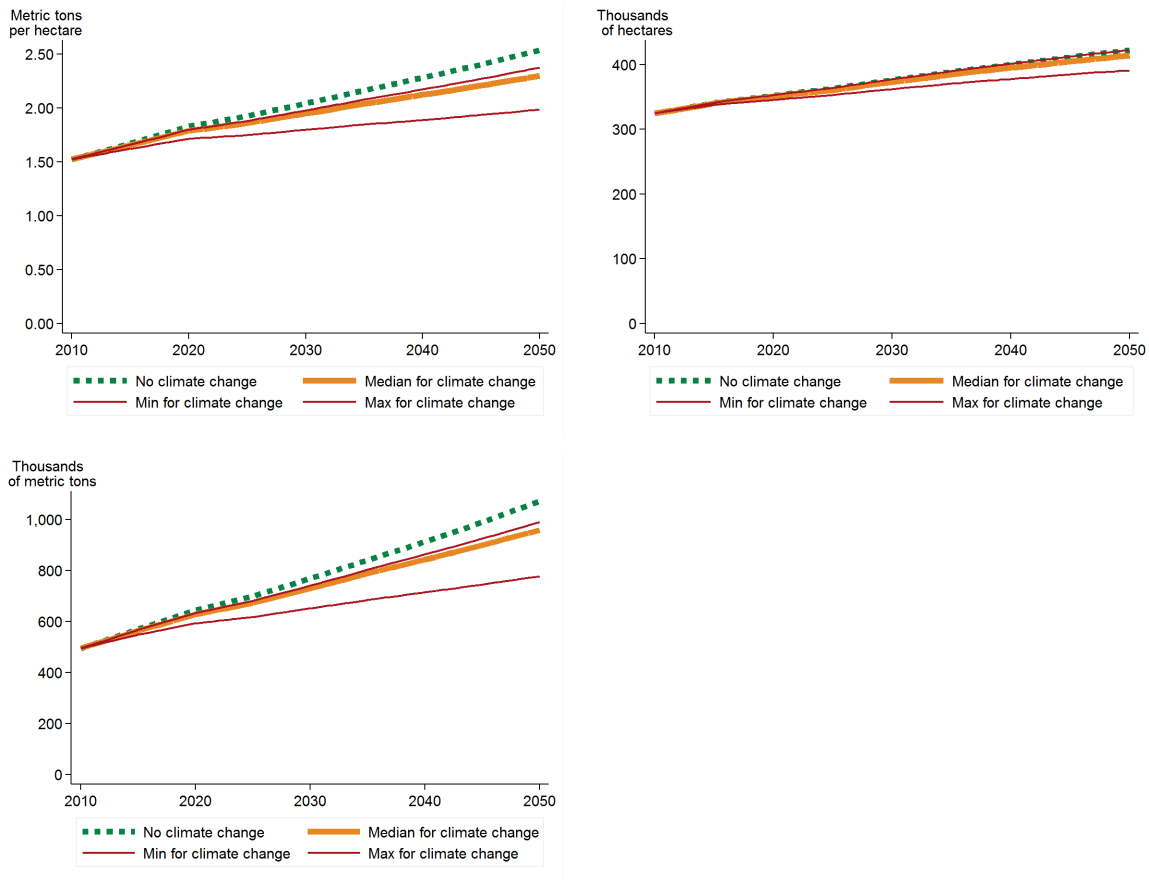
Figure 24 Median yield change (percent) under climate change: rainfed maize in Honduras, 2000–2050



Source: Authors' calculations based on data from the Agricultural Model Intercomparison and Improvement Project (AgMIP) Global Gridded Crop Model Intercomparison (GGCMI; see Rosenzweig et al. 2014).

Note: The median at each pixel was computed from all possible combinations of four crop models and four climate models. All are from AGMIP GGCMi for Representative Concentration Pathway 8.5 (RCP8.5)

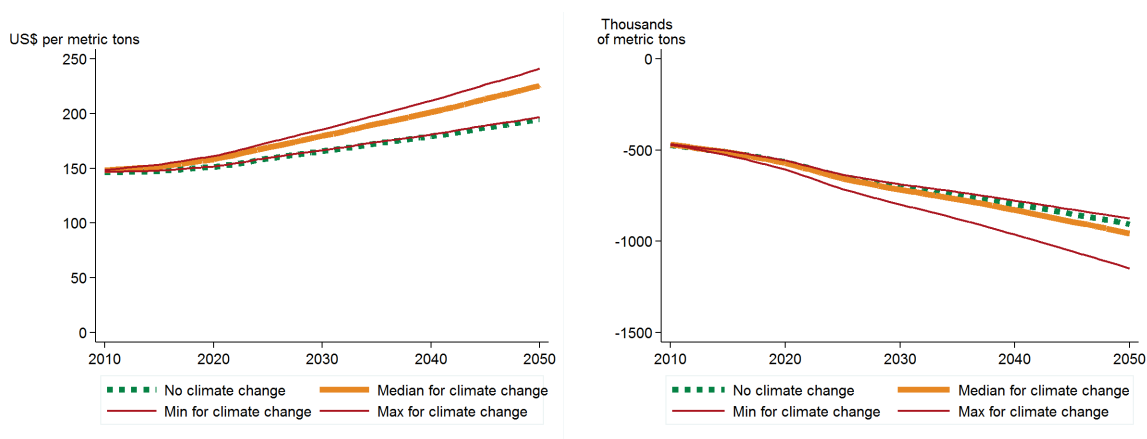
Figure 25 Yield, harvested area, and production projections from the IMPACT model: maize in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

Note: Yield, top left; harvested area, top right; production, bottom left.

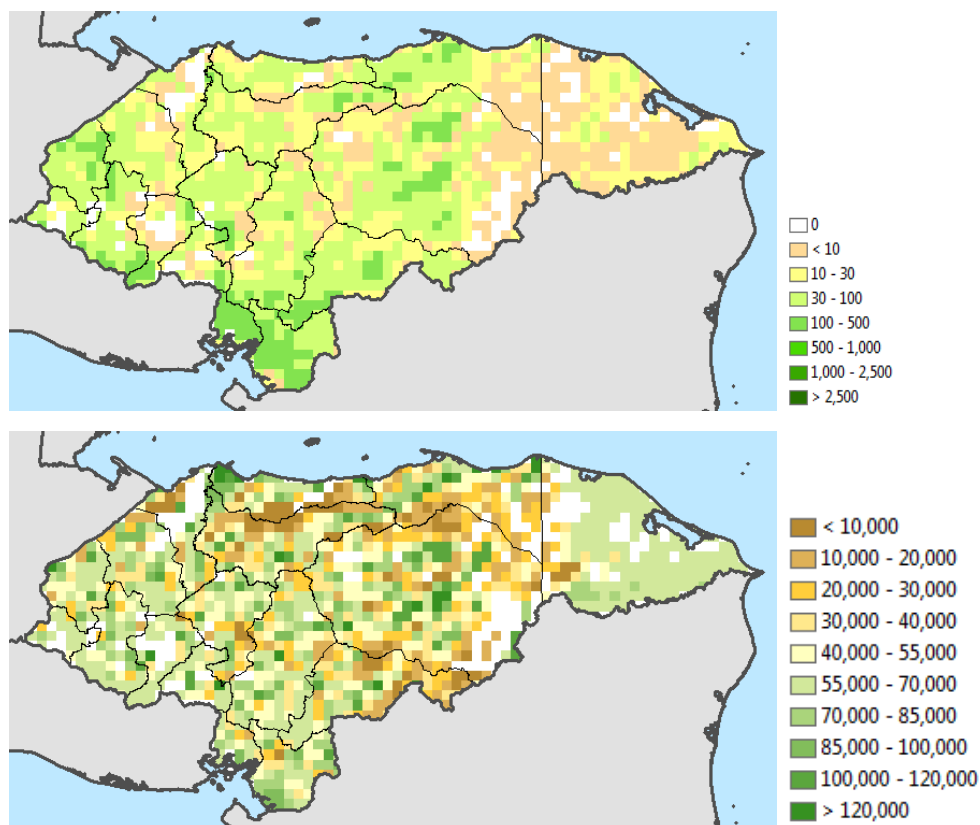
Figure 26 World price and net exports from the IMPACT model: maize in Honduras, 2010–2050



Source: Authors/IFPRI, using Representative Concentration Pathways version 8.5 (RCP8.5).

Notes: World price, top left; net exports, top right.

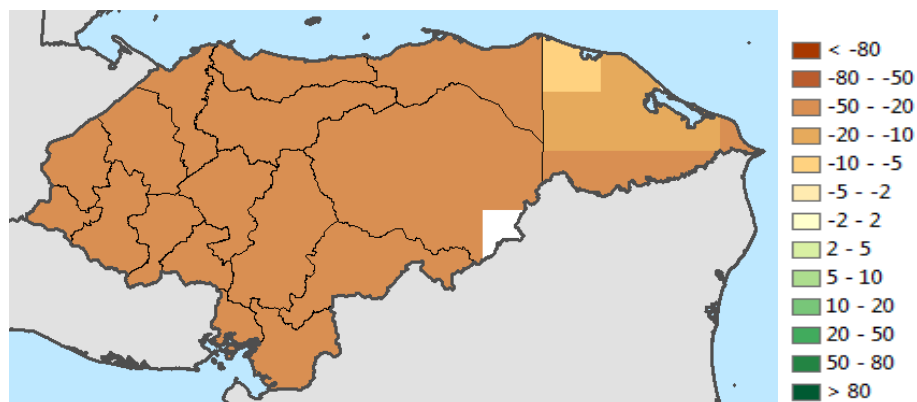
Figure 27 Intensity (hectares per pixel) and productivity of rainfed sugarcane in Honduras, circa 2005



Source: Spatial Production Allocation Model (SPAM, You et al. 2014).

Note: A pixel at the equator contains approximately 8,500 hectares.

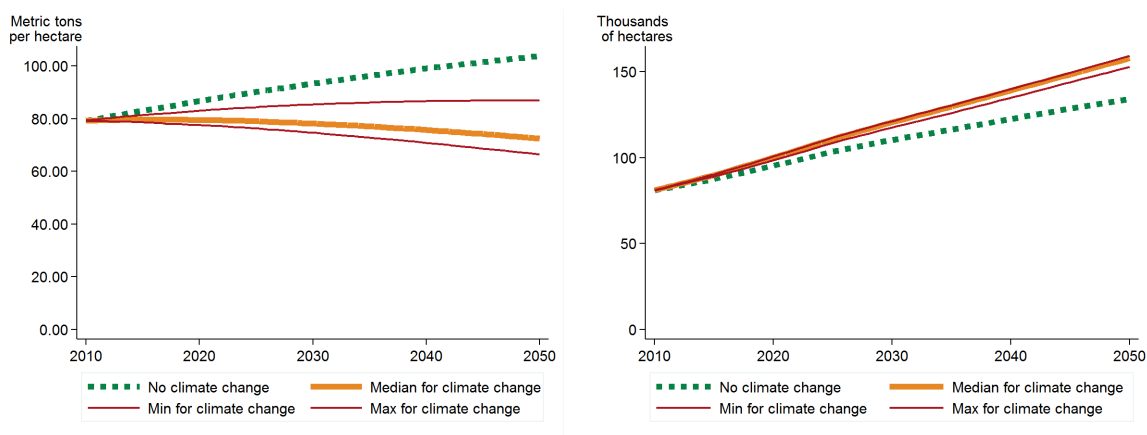
Figure 28 Median yield change (percent) under climate change: rainfed sugarcane in Honduras, 2010–2050

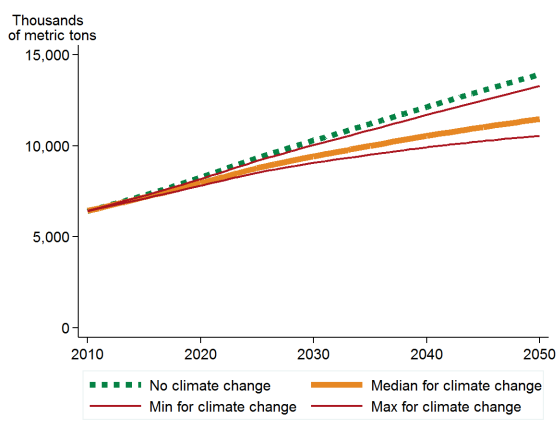


Source: Authors' calculations based on data from the Agricultural Model Intercomparison and Improvement Project (AgMIP) Global Gridded Crop Model Intercomparison (GGCMI; see Rosenzweig et al. 2014).

Note: There were a total of one crop model and four climate models. All are from the Coupled Model Intercomparison Project (CMIP) for the Agricultural Model Intercomparison and Improvement Project (AgMIP), and are for Representative Concentration Pathways, version 8.5 (RCP8.5).

Figure 29 Yield, harvested area, and production projections from the IMPACT model: sugarcane in Honduras, 2010–2050

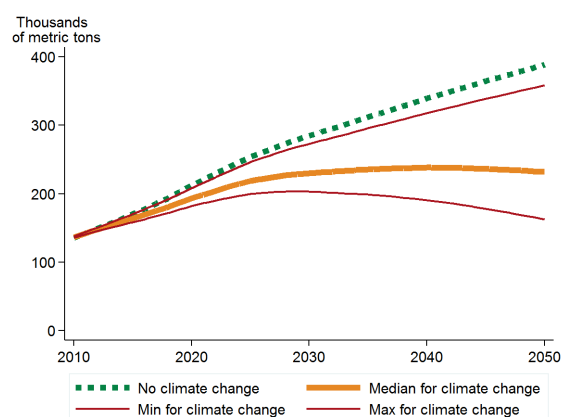
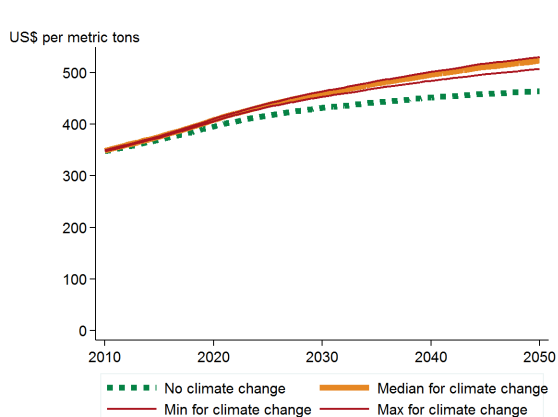




Source: Authors, using the IMPACT model (Robinson et al, 2015).

Note: Yield, top left; harvested area, top right; production, bottom left.

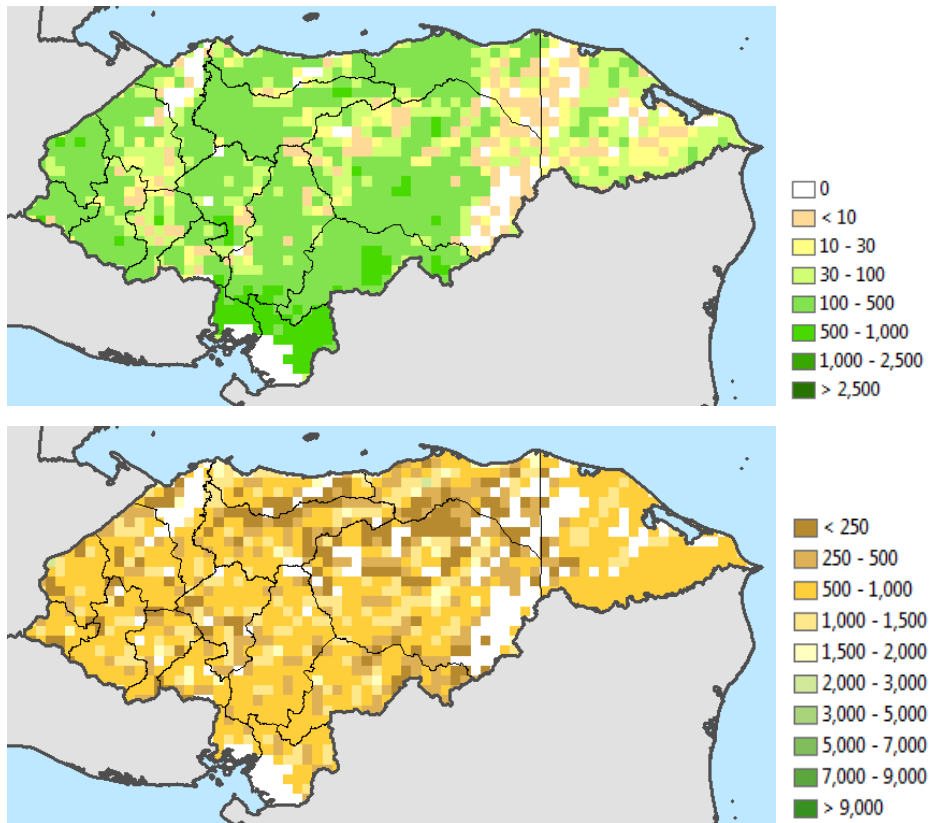
Figure 30 World price and net exports from the IMPACT model: sugar in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

Note: World price, top left; net exports, top right.

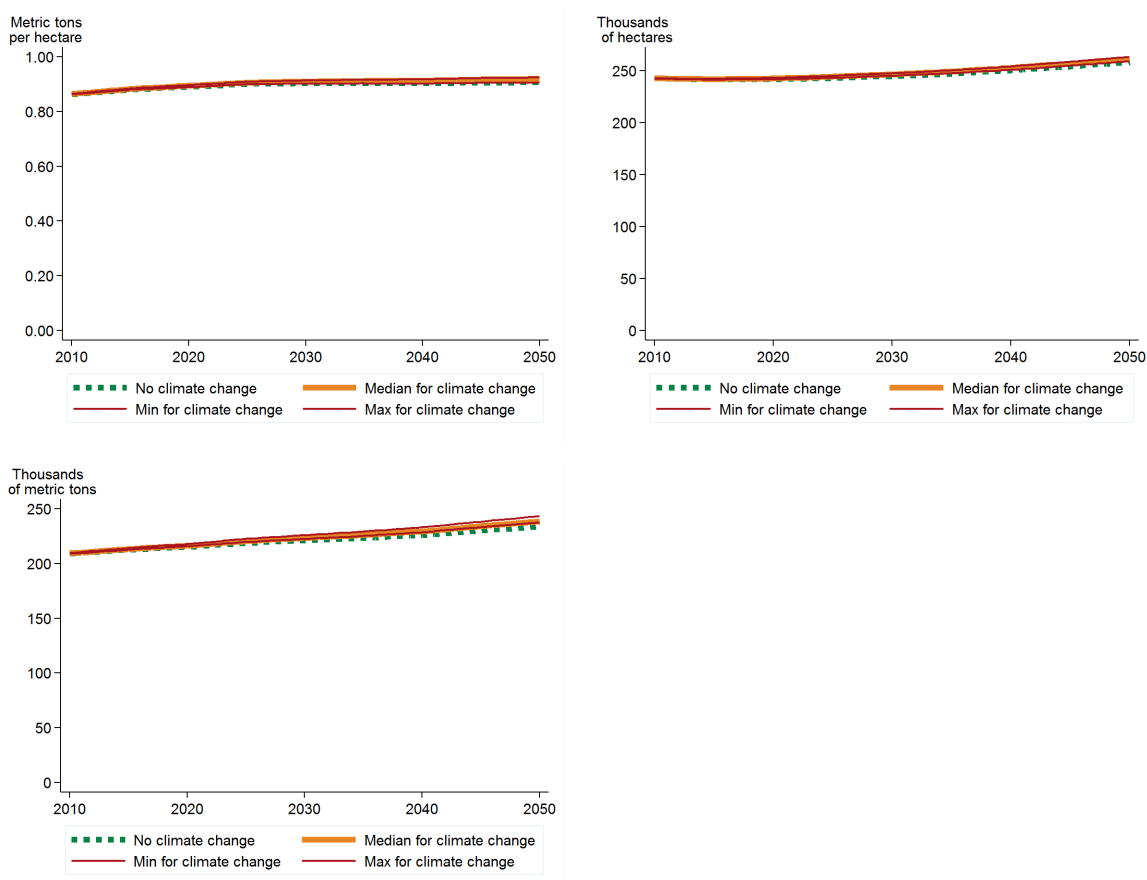
Figure 31 Intensity (hectares per pixel) and productivity (kilograms per hectare) of rainfed coffee in Honduras, circa 2005



Source: Spatial Production Allocation Model (SPAM, You et al. 2014).

Note: A pixel at the equator contains approximately 8,500 hectares.

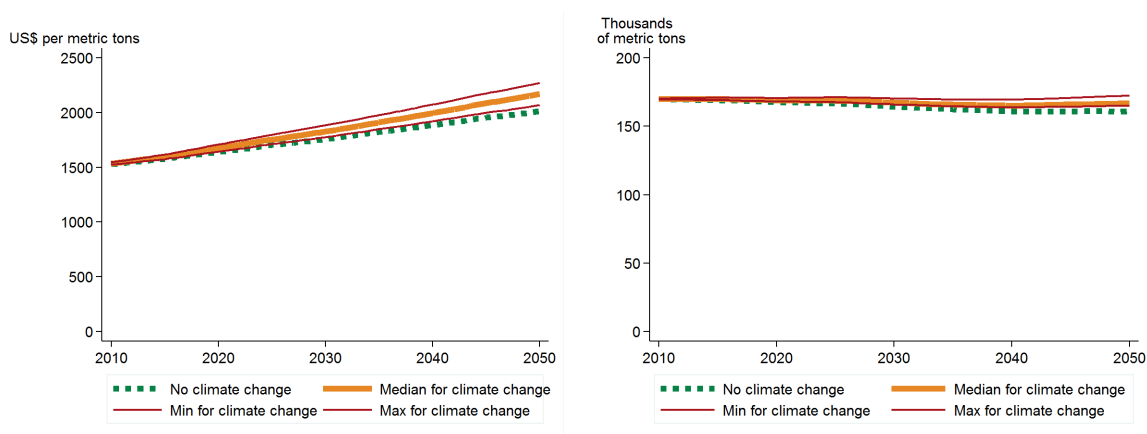
Figure 32 Yield, harvested area, and production projections from the IMPACT model: coffee in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

Note: Yield, top left; harvested area, top right; production, bottom left.

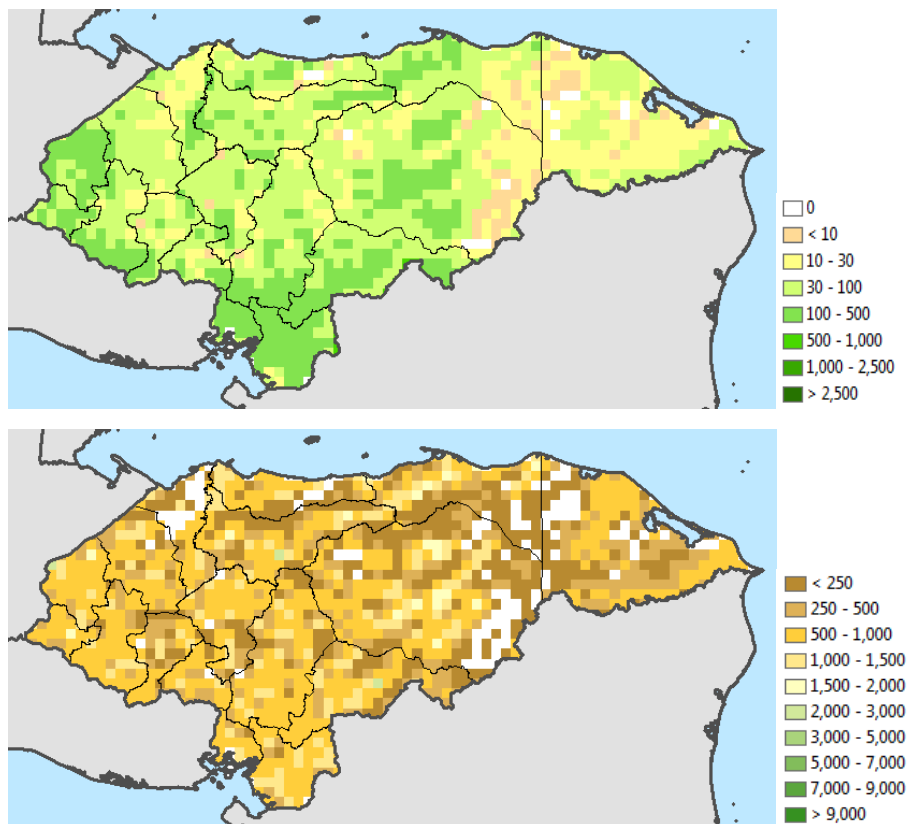
Figure 33 World price and net exports from the IMPACT model: coffee in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

Note: World price, top left; net exports, top right.

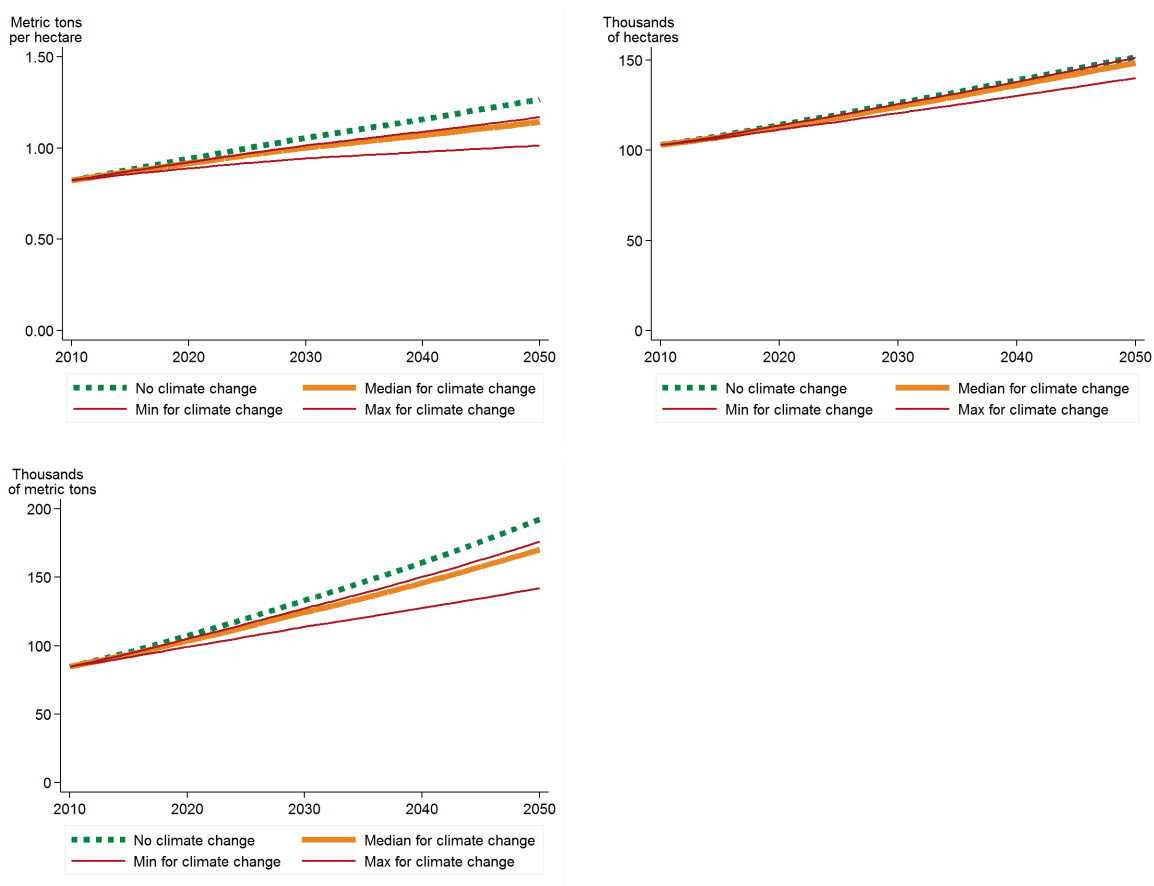
Figure 34 Intensity (hectares per pixel) and productivity of rainfed beans in Honduras, circa 2005



Source: Spatial Production Allocation Model (SPAM, You et al. 2014).

Note: A pixel at the equator contains approximately 8,500 hectares.

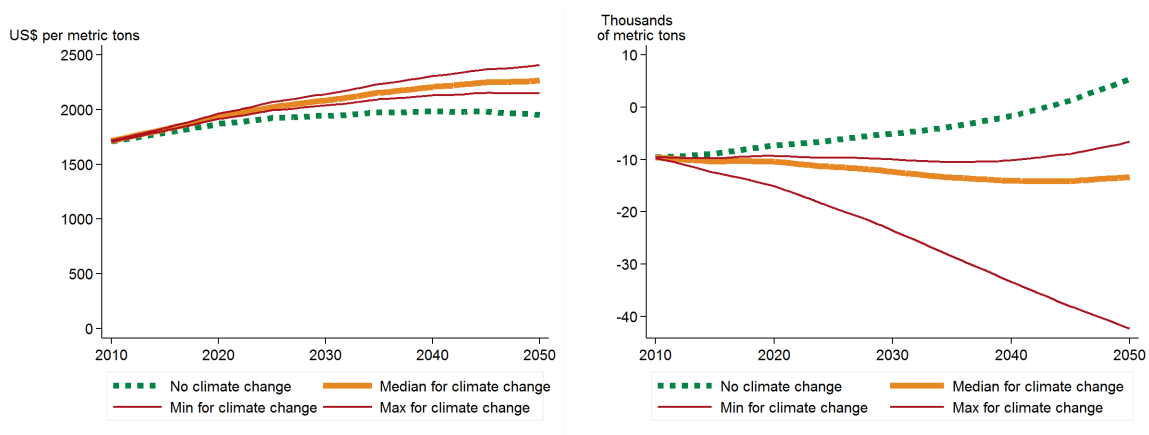
Figure 35 Yield, harvested area, and production projections from the IMPACT model: beans in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

Note: Yield, top left; harvested area, top right; production, bottom left.

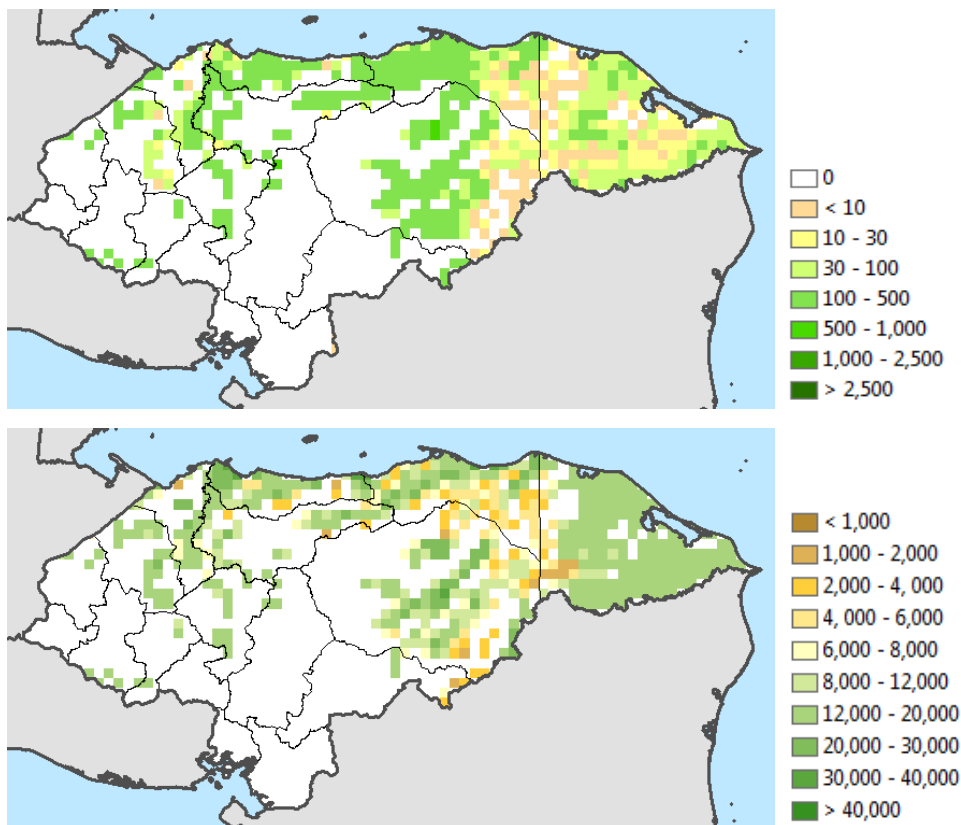
Figure 36 World price and net exports from the IMPACT model: beans in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

Note: World price, top left; net exports, top right.

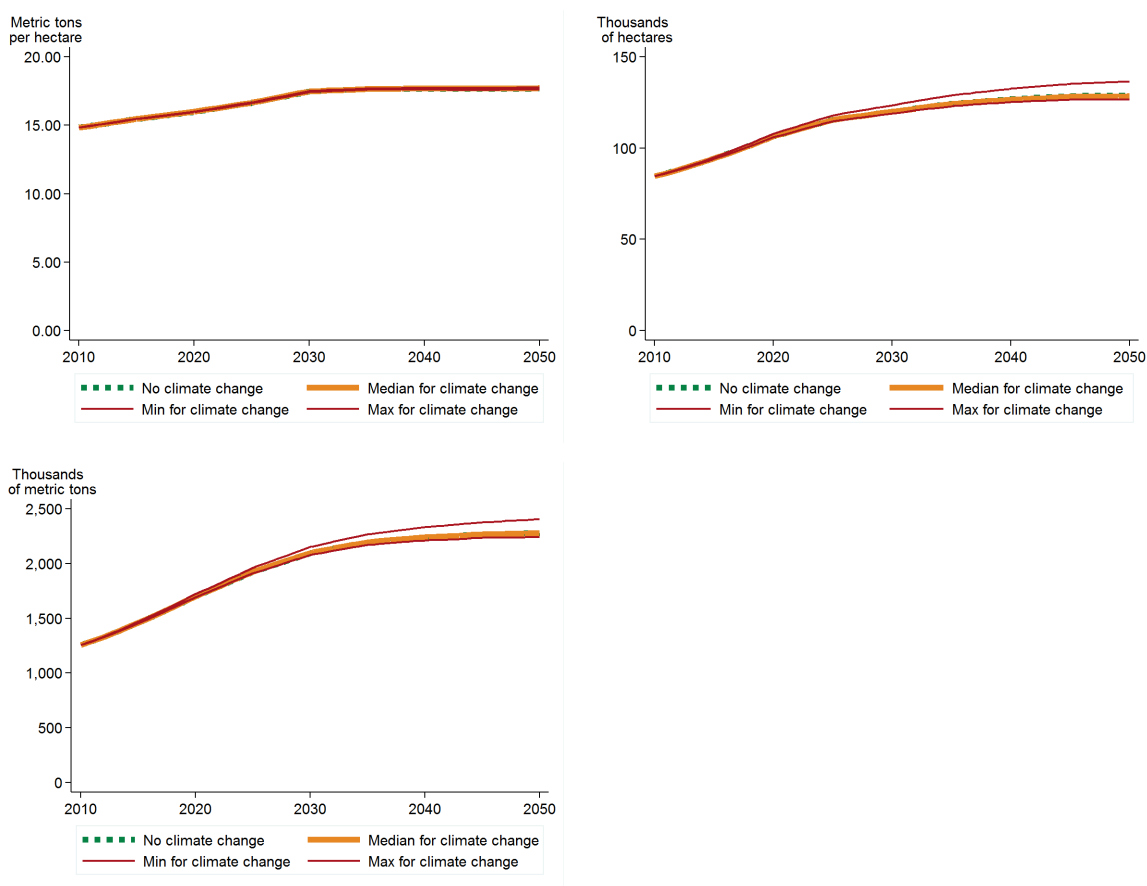
Figure 37 Intensity (hectares per pixel) and productivity of palm oil in Honduras, circa 2005



Source: Spatial Production Allocation Model (SPAM, You et al. 2014).

Note: A pixel at the equator contains approximately 8,500 hectares.

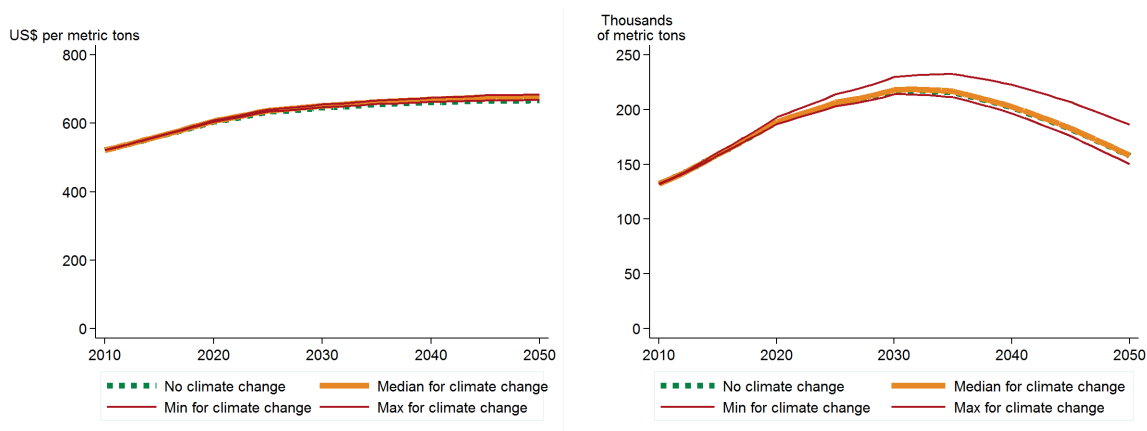
Figure 38 Yield, harvested area, and production projections from the IMPACT model: palm oil in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

Note: Yield, top left; harvested area, top right; production, bottom left.

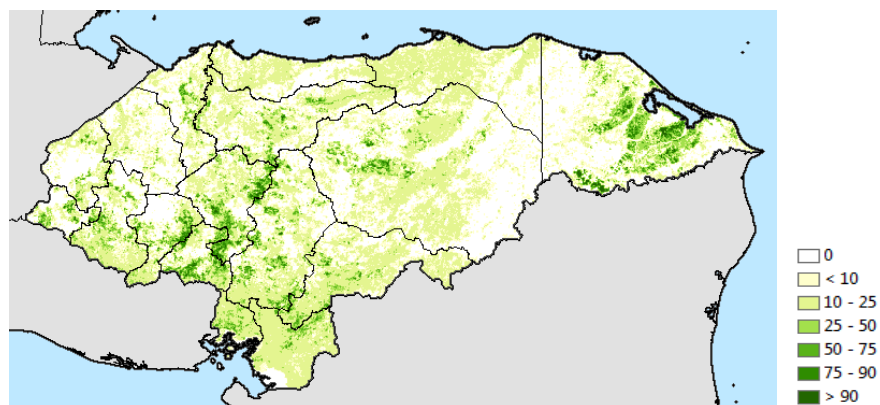
Figure 39 World price and net exports from the IMPACT model: palm oil in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

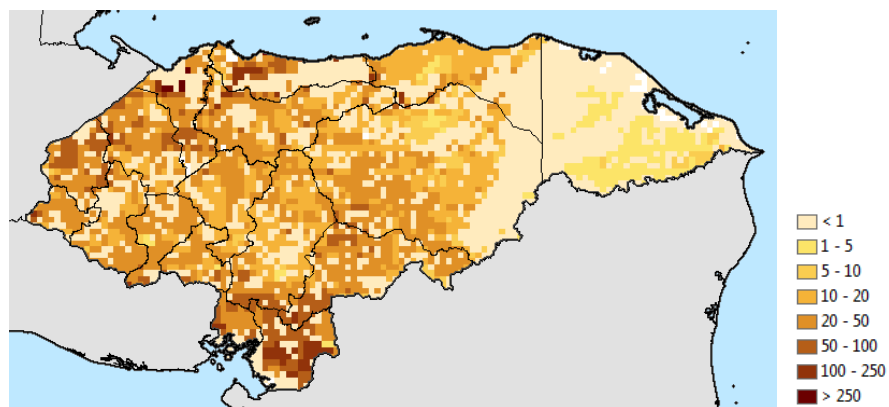
Note: World price, top left; net exports, top right.

Figure 40 Percentage of grassland in Honduras in 2010



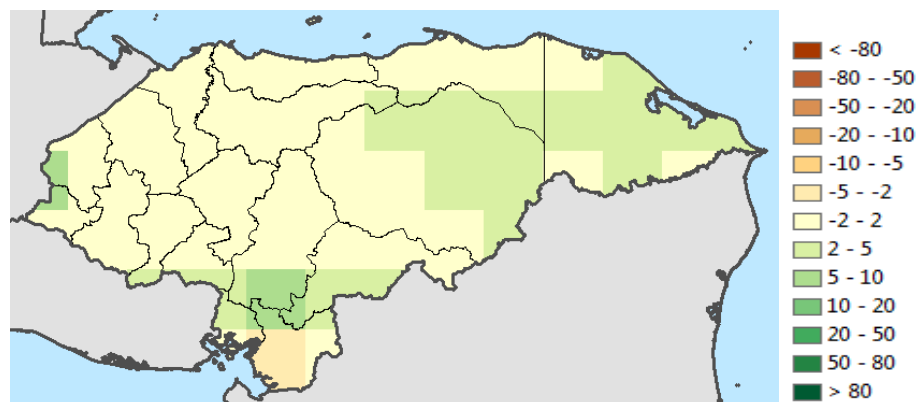
Source: GLC-SHARE (Latham et al. 2014).

Figure 41 Cattle density map of Honduras (head per square kilometer)



Source: Gridded Livestock of the World (FAO 2007).

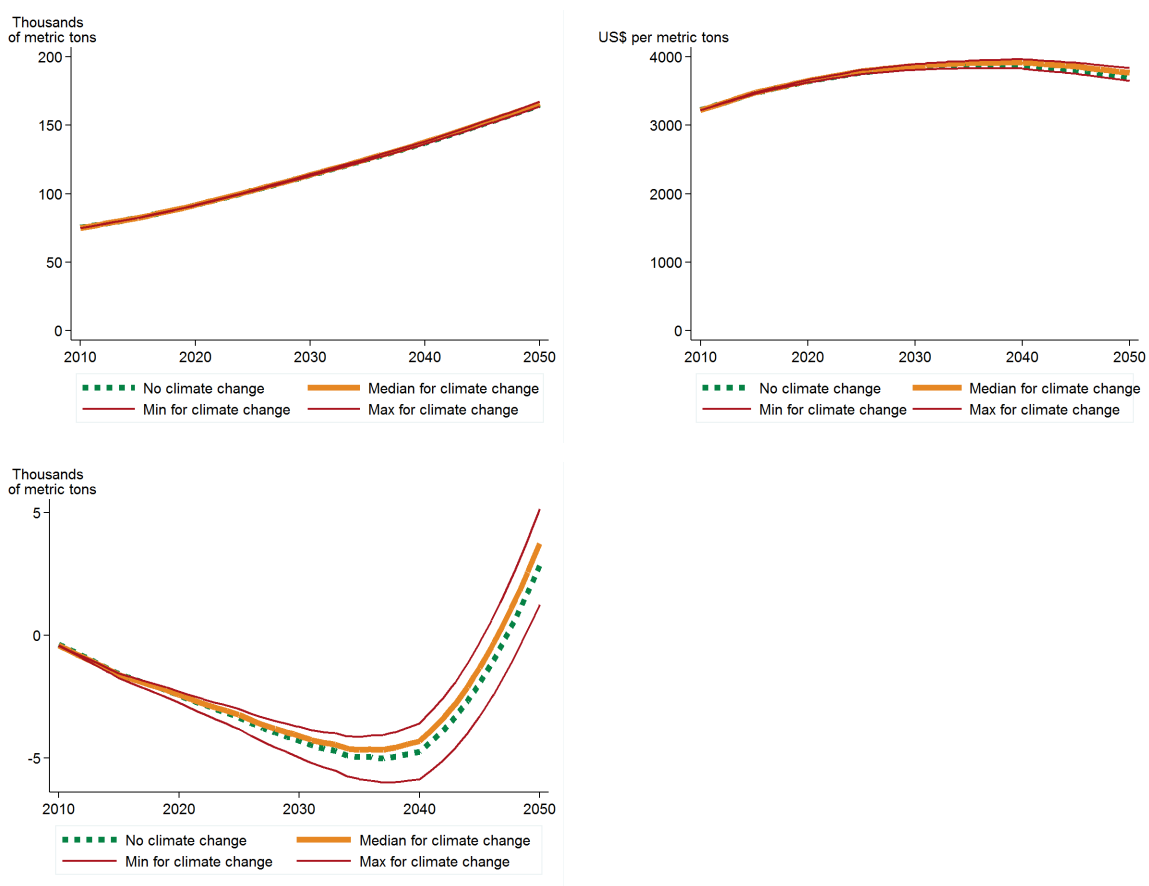
Figure 42 Median yield change (percent) under climate change: rainfed managed grasses in Honduras, 2000–2050



Source: Authors' calculations based on data from the Agricultural Model Intercomparison and Improvement Project (AgMIP) Global Gridded Crop Model Intercomparison (GGCMI; see Rosenzweig et al. 2014).

Note: These results reflect one biophysical model and four climate models. All are from the Coupled Model Intercomparison Project (CMIP) for the Agricultural Model Intercomparison and Improvement Project (AgMIP), and are for Representative Concentration Pathways, version 8.5 (RCP8.5).

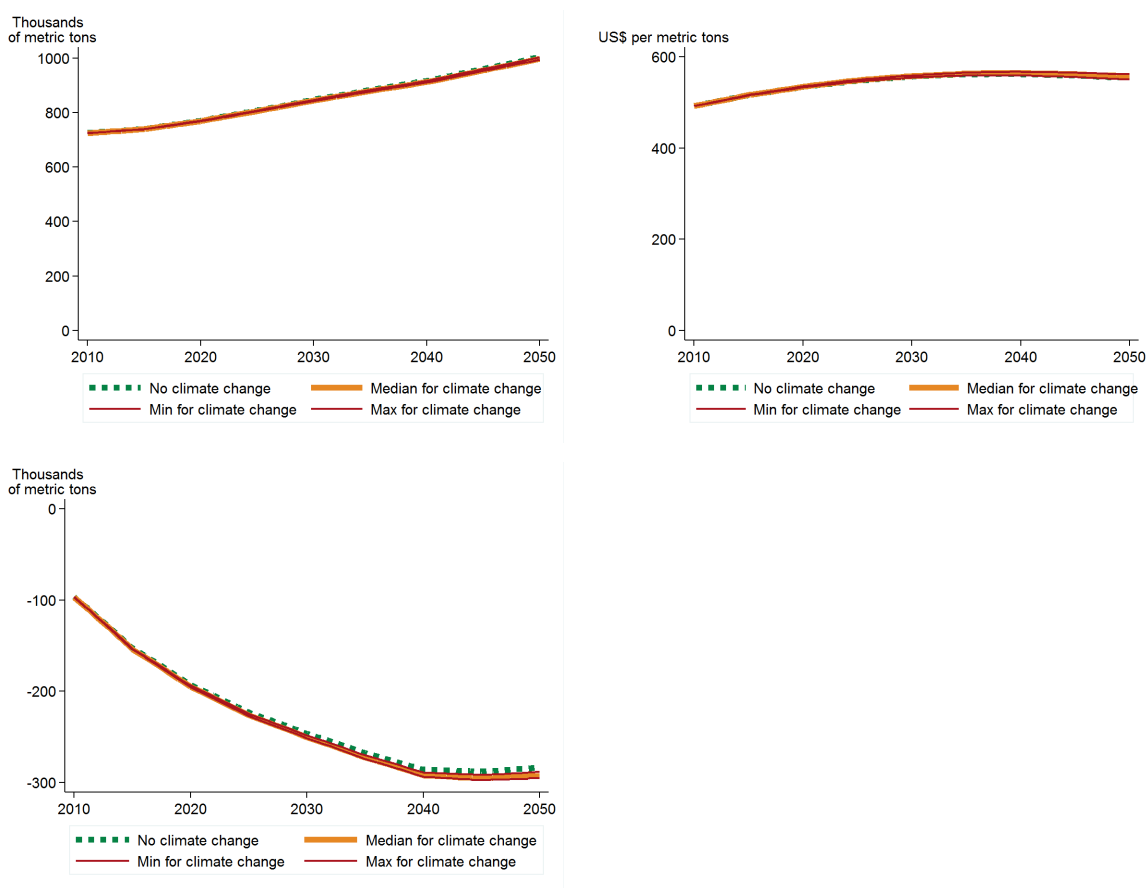
Figure 43 Production, world price, and net exports from the IMPACT model: beef in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

Note: Production, top left; world price, top right; net exports, bottom left

Figure 44 Production, world price, and net exports from the IMPACT model: milk in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

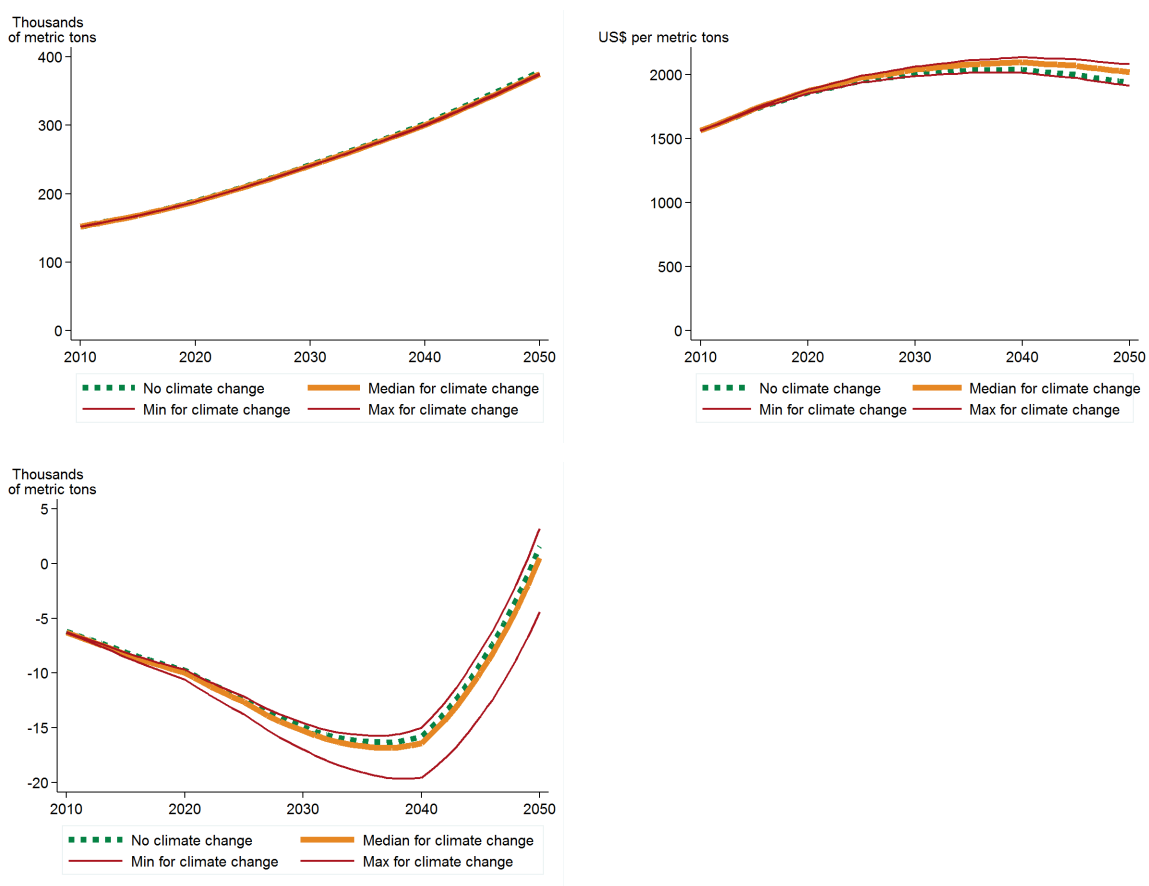
Note: Production, top left; world price, top right; net exports, bottom left.

Figure 45 Chicken density map of Honduras



Source: Gridded Livestock of the World (FAO 2007).

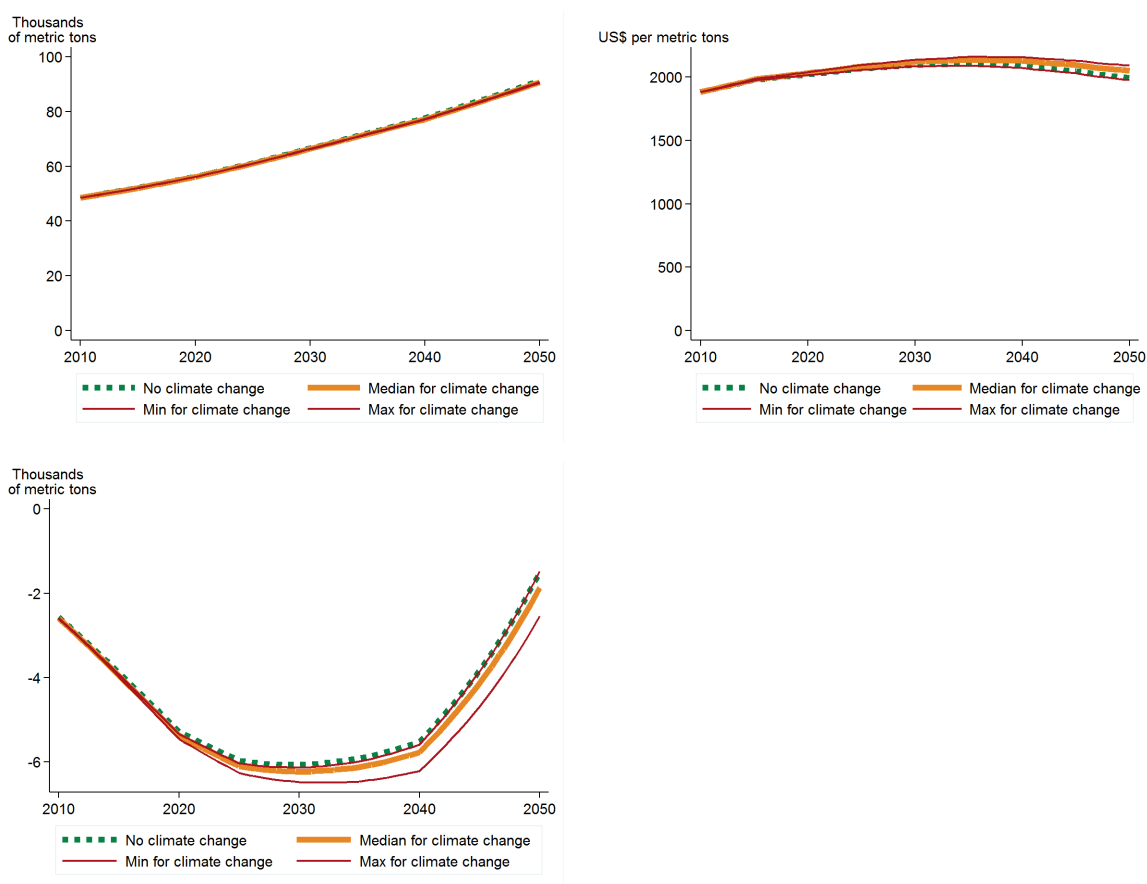
Figure 46 Production, world price, and net exports from the IMPACT model: chicken meat in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

Note: Production, top left; world price, top right; net exports, bottom left.

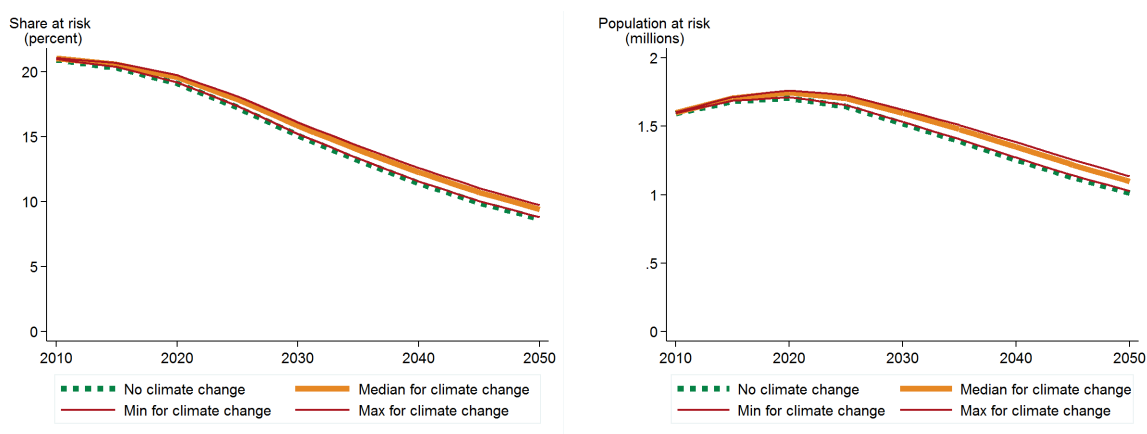
Figure 47 Production, world price, and net exports from the IMPACT model: chicken eggs in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

Note: Production, top left; world price, top right; net exports, bottom right.

Figure 48 Share of population and total number of people at risk of hunger in Honduras, 2010–2050



Source: Authors, using the IMPACT model (Robinson et al, 2015).

TABLES

Table 1 Education and nutrition statistics for Honduras, 2010–2012

Indicator	Year	Percent
Literacy rate, adult total (% of people ages 15 and above)	2011	85.1
School enrollment, primary (% of gross, three-year average)	2012	112.8
School enrollment, secondary (% of gross, three-year average)	2012	73.1
Employment in agriculture (% of total employment, three-year average)	2011	36.1
Vulnerable employment, total (% of total employment, three-year average)	2010	51.6
Malnutrition prevalence, weight for age (% of children under 5)	2012	7.1
Access to electricity (% of population)	2011	83.3

Source: World Development Indicators (World Bank 2014).

Table 2 Climate statistics, 1960–1990

Region	Rain (millimeters)			Mean daily maximum temperature for warmest month (°C)
	Annual	Driest 3 months	Wettest 3 months	
Honduras	1,734	129	719	31.49

Source: WorldClim version 1.4 (Hijmans et al. 2005).

Table 3 Summary of climate changes from general circulation models, 2000–2050

Change in annual precipitation (millimeters)				Change in mean daily maximum temperature for warmest month (°C)			
GFDL	HadGEM	IPSL	MIROC	GFDL	HadGEM	IPSL	MIROC
521	–98	–352	–181	1.89	2.78	2.26	2.65

Source: CMIP5.

Note: GFDL = Geophysical Fluid Dynamics Laboratory; HadGEM = Hadley Centre Global Environmental Model; IPSL = L’Institut Pierre-Simon Laplace; MIROC = Model for Interdisciplinary Research on Climate.

Table 4 Harvested area, production, and yield of leading agricultural commodities in Honduras, mean of 2010–2012, with changes from mean of 1999–2001

Item	Rank by area harvested	Hectares harvested, mean for 2010–2012	% change 2001–2012, hectares harvested	Tons produced, mean for 2010–2012	% change 2001–2012, tons produced	Yield (tons/hectare), 2010–2012	% change, yield
Maize	1	440,941	19.64	563,987	10.79	1.28	–7.25
Coffee, green	2	267,064	26.68	270,576	45.93	1.01	14.77
Beans, dry	3	130,808	27.11	86,564	31.40	0.66	3.13
Oil, palm fruit	4	110,667	181.36	1,411,749	123.28	12.76	–20.60
Sugarcane	5	76,516	62.99	8,029,974	103.35	104.95	24.76
Sorghum	6	50,705	–29.84	46,811	–33.33	0.92	–5.15
Bananas	7	25,525	13.69	757,051	58.00	29.66	38.99
Oranges	8	17,858	31.89	277,098	114.16	15.52	62.34
Rice, paddy	9	10,127	151.98	43,502	322.64	4.3	67.97
Melons, other (including cantaloupes)	10	9,033	58.95	266,667	171.64	29.52	70.93

Source: FAOSTAT (FAO 2014).

Note: Percentage change is comparing the three-year average for 1999–2001 with the three-year average for 2010–2012.

Table 5 Livestock in Honduras, mean of 2010–2012

Item	Quantity	Units
Cattle	2,668,294	Head
Pigs	475,878	Head
Horses	181,267	Head
Mules	69,817	Head
Chickens	40,030	Thousand head
Goats	24,833	Head
Asses	23,133	Head
Sheep	15,833	Head
Beehives	10,633	Number

Source: FAOSTAT (FAO 2014).

Table 6 Gross value of production of leading agricultural commodities in Honduras, 2010–2012

Item	Rank by gross value of production	Gross value of production, avg. 2010–2012
Coffee, green	1	290,696
Sugarcane	2	263,680
Milk, whole fresh cow	3	249,061
Meat indigenous, chicken	4	221,366
Bananas	5	213,209
Meat indigenous, cattle	6	164,418
Oil, palm	7	143,569
Maize	8	79,897
Tomatoes	9	57,409
Oranges	10	53,551

Source: FAOSTAT (FAO 2014).

Note: Value of production is for the three-year average for 2010–2012. Value is in thousands of constant international dollars, which holds the price to the 2004–2006 value. "The method assigns a single price to each commodity and country. For example, 1 ton of maize has the same price in whichever country it is produced" (FAO 2014).

Table 7 Value of leading exports of agricultural commodities in Honduras, 2009–2011

Item	Rank by value	Value of exports, avg. 2009–2011
Coffee, green	1	865,461
Bananas	2	304,323
Oil, palm	3	166,473
Cigars, cheroots	4	65,640
Melons, other (including cantaloupes)	5	42,359
Sugar raw centrifugal	6	32,407
Cigarettes	7	31,652
Pineapples	8	24,719
Pastry	9	23,066
Oil, palm kernel	10	19,540

Source: FAOSTAT (FAO 2014).

Note: Value is the three-year average for 2009–2011. Value is in thousands of constant international dollars, which holds the price to the 2004–2006 value.

Table 8 Value of leading imports of agricultural commodities in Honduras, 2009–2011

Item	Rank by value	Value of imports, avg. 2009–2011
Maize	1	104,557
Soybean cake	2	66,120
Pastry	3	64,566
Wheat	4	63,004
Rice—total	5	54,303
Nonalcoholic beverages	6	46,959
Breakfast cereals	7	34,490
Food wastes	8	31,210
Infant food	9	26,107
Sugar confectionery	10	25,363

Source: FAOSTAT (FAO 2014).

Note: Value is the three-year average for 2009–2011. Value is in thousands of constant international dollars, which holds the price to the 2004–2006 value.

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